Lexical Task Heads: Identifying Them on Simple Tasks with Logit Lens and Jacobian Lens Readouts

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Abstract

Lexical task heads are attention heads whose output, decoded into the vocabulary, names the task the prompt asks for (Yang et al., 2026). They were identified with the per-head logit lens on simple single-step tasks, such as mapping a country to its capital. We work on how they are identified, on 14 of these tasks in Llama-3.1-8B-Instruct: whether the method of Yang et al. (2026) reproduces, whether Head-JLens (Merlo, 2026), which decodes each head's output after mapping it through a Jacobian of the remaining layers averaged over many texts, finds lexical task heads that the logit lens misses, and which heads to keep when a task is asked for in several wordings. Heads are identified under both readouts with rule-based and example-based prompts and tested by activation patching. A companion document applies the method to complex, rule-defined tasks.

1 Introduction

A language model's accuracy on a task can change a great deal with the wording of the prompt that asks for it: on Llama-3.1-8B-Instruct, prompts for the same antonym items differ in accuracy by up to a factor of four (Yang et al., 2026, Fig. 1). Yang et al. (2026) explain part of this variation with *lexical task heads*, attention heads whose outputs, decoded into the vocabulary with the logit lens, name the task ("opposite" for antonyms, "cities" for capitals). The heads are largely shared between instruction-based and example-based prompts, patching their mean activation into a prompt without task information raises task accuracy, and prompts that fail activate them less.

The identification step leaves open choices that this document works on. First, the readout: the logit lens decodes a head's output directly and leaves out the computation of the layers that follow it; Yang et al. (2026, Sec. 4) note that task representations "encoded in a non-lexical, abstract format" may not project cleanly to tokens. Head-JLens reads each head through a Jacobian map of the remaining layers, and on Llama-3.1-8B-Instruct it ranks annotated intermediate words, which the prompt never states (such as the bridge entity of a two-hop question), higher than the per-head logit lens at most layers on five of six evaluation sets (Merlo, 2026, Sec. 3). Second, the prompts the heads are read from: the code of Yang et al. (2026) reads the rule-style heads from one instruction template, the most accurate, although five are used for each task.

We therefore ask whether the method of Yang et al. (2026) reproduces on our setup, how the logit lens and Head-JLens compare as readouts for identifying lexical task heads, and which heads should be kept when a task is asked for in several wordings. A companion document (`proposal-complex.tex`) applies the method to the complex, rule-defined tasks of Fu et al. (2026).

Contributions.

1. A replication of the lexical-task-head method of Yang et al. (2026) on Llama-3.1-8B-Instruct, with both causal checks their paper describes: patching prompts without task information (their App. G.1), and patching prompts answered wrongly (their Sec. 2.4.1 and App. H.1.3) in the setting of their released script.

2. A comparison of the per-head logit lens and Head-JLens for identifying lexical task heads, with a causal check of the heads each finds.

3. A comparison of rules for selecting heads across instruction templates.

Scope. We do not explain behavioural variation across wordings (the failure analyses of Yang et al., 2026, Sec. 2.4), and we use one model.

Organisation. After the shared framework and methods (Secs. 3 and 4), Sec. 5 replicates Yang et al. (2026) under both readouts, and each later section tests one change to how heads are selected: its design, followed by its results.

2 Related work

Task representations across prompting styles. Function vectors are sums of the mean outputs of a few attention heads that, added to the residual stream, trigger a task in another context (Todd et al., 2024). Davidson et al. (2025) extract them from instructions as well as from demonstrations and find that the two prompting styles engage mostly different heads: of the 20 top heads in each model, 2 to 7 are shared. Yang et al. (2026) identify a different kind of task representation, lexical task heads, whose outputs decode to words describing the task; 73% of them are shared between instruction-based and example-based prompts on average, and they are largely disjoint from function-vector heads (Yang et al., 2026, App. I). We work on how lexical task heads are identified.

Reading attention heads in the vocabulary. The logit lens decodes an intermediate state with the model's final normalization and unembedding (nostalgebraist, 2020); applied to one head's output, it is the readout of Yang et al. (2026). The Jacobian lens of Gurnee et al. (2026) instead maps a residual-stream state through a Jacobian of the remaining layers, averaged over many texts. Head-JLens moves this map to the concatenated head outputs, so that each head is read through the layers that follow it; in Llama-3.1-8B-Instruct it ranks annotated intermediate words higher than the per-head logit lens at most layers on five of six evaluation sets, and heads selected with it are better sites for edits (Merlo, 2026). We compare the two readouts on a different use: identifying task-descriptive heads.

3 Framework

Tasks and prompts. A task t is given by a dataset D_t of inputs and correct outputs. Each prompt for t has a style $s \in \{\text{rule}, \text{example}\}$: its task information is an instruction or a set of demonstrations, followed by the input x. $P_{t,s}$ is the set of prompts of style s for t, and $P^+_{t,s} \subseteq P_{t,s}$ the prompts whose heads are read: those the model answers correctly under the flexible score of Sec. 4.5, whether or not the answer is the first generated text.

Heads and readouts. Following Merlo (2026, Sec. 2), $z_{l,h}(p)$ is the output of head h in layer l at the last position of p, before the output projection, and $c_{l,h}(p) = W^{(h)}_{O,l} z_{l,h}(p)$ its write to

the residual stream. With $q_v = g \odot w_v$ the gain-scaled unembedding row of token v, the two readouts score v by

$$\ell^{\mathrm{L}}_{l,h}(p)[v] = q_v^\top c_{l,h}(p) \quad \text{(Logit-head)}, \qquad \ell^{\mathrm{J}}_{l,h}(p)[v] = q_v^\top J_{l,h}\, z_{l,h}(p) \quad \text{(J-head)}.$$

$T^R_{k,l,h}(p)$ is the set of the k highest-scoring tokens under readout $R \in \{\mathrm{L}, \mathrm{J}\}$, and V_t the term list of task t.

Definition 1 (Lexical task head)**.** *Following Yang et al. (2026, Sec. 2.2.2), head (l, h) is a lexical task head for t on prompt p under readout R if $|T^R_{k,l,h}(p) \cap V_t| \geq n$. It is a lexical task head for t and style s if this holds for at least a fraction ρ of the prompts in $P^+_{t,s}$. We use $n = 1$, $k = 10$, $\rho = 0.1$, as Yang et al. (2026). $H^R_{t,s}$ denotes the set of such heads.*

4 Methods

4.1 Model

Llama-3.1-8B-Instruct: it is the main model of Yang et al. (2026), and the only model with released Head-JLens matrices.

4.2 Tasks

The 14 tasks used by both Yang et al. (2026) and Davidson et al. (2025), with the datasets of Todd et al. (2024): antonym, synonym, singular–plural, present–past, country–capital, country–currency, landmark–country, park–country, product–company, next-item, prev-item, English–French, English–German, English–Spanish. Rule-based prompts: the first five of the six instruction templates per task listed in Yang et al. (2026, App. A.3.1), that is the cloze form ("The antonym of {x} is") and four question–answer forms; their code and figures use only these five (templates A–E), not the sixth. Example-based prompts: 1, 5, 10, 20 and 30 shots (Yang et al., 2026, App. A.3).

Why these tasks. Lexical task heads and their term lists were defined on these tasks (Yang et al., 2026), so the method applies unchanged and our results can be compared with theirs.

Answer format. As in Yang et al. (2026): their prompt templates, as raw text without the chat template. Decoding is greedy, so that the prompt whose heads are read and the scored answer always match. Lexical task heads are read at the last prompt token, before any generated text.

4.3 Task-representation methods

Lexical task heads (Yang et al., 2026, Sec. 2.2.2). Each attention head's output at the last prompt token is decoded with the two readouts of Sec. 3: Logit-head, the per-head logit lens used by Yang et al. (2026) (nostalgebraist, 2020), and J-head (Head-JLens), which applies the head's columns $J_{l,h}$ of the per-layer Jacobian map fitted on WikiText-103 (Merlo, 2026, Sec. 2). Fitted matrices for Llama-3.1-8B-Instruct are released with Merlo (2026, Sec. S4). Under each readout, lexical task heads are identified with Definition 1. $P^+_{t,s}$ is taken as in the code of Yang et al. (2026): the first 100 prompts in P^+ under one wording, the 5-shot prompts for the example style and the instruction template with the highest accuracy for the rule style. Term lists: Yang et al. (2026, App. A.2.1).

Simple tasks as in the source code. The simple tasks are run as in the code of Yang et al. (2026), with three departures. First, padding: their code passes prompts to the model in batches of 20, padded on the left, without an attention mask, so that the model attends to the padding; we pass the mask and the matching positions, so that each prompt is computed as if run alone. Second, the J-head readout. Third, the precision of the forward pass: bfloat16, in which the Head-JLens matrices were fitted.

Why these readouts. Lexical task heads locate a task representation and give its content in words (Yang et al., 2026). Logit-head reproduces their readout. J-head reads each head through the layers that follow it: in Llama-3.1-8B-Instruct it ranks unstated intermediate words higher than Logit-head at most layers on five of six evaluation sets, and its readouts are better guides for edits (Merlo, 2026, Secs. 3–4). A lexical task head's decoded task word is likewise not the answer, and Yang et al. (2026, Sec. 4) note that their logit-lens criterion may miss task representations that do not project cleanly to tokens.

4.4 Causal check

The causal check of Yang et al. (2026, Sec. 2.3.2, App. G) tests whether a head set carries the task: the heads' mean output on task prompts is written into prompts that do not state the task, and we measure whether the model then performs it.

1. *Patched prompts*: prompts without task information, the zero-shot prompts of Yang et al. (2026), for items that the model answers correctly when the task is given. Their unpatched accuracy is the baseline.

2. *Mean output*: for each head in the set, the mean of $c_{l,h}$ over the first 100 prompts of a wording answered correctly, leaving out the patched items. As in the code of Yang et al. (2026), each wording of a style (each instruction template, each number of shots) gives its own mean and its own patch.

3. *Patch*: at the last position of each patched prompt, the output of every head in the set is replaced by its mean output times a scaling factor $\alpha \in \{-10, -6, -4, -2, 0, 1, 2, \ldots, 10\}$, the values of Yang et al. (2026). The model answers from the patched prompt, generated as the saved outputs of Sec. 4.5, and the answer is scored by the flexible score. This replaces the first generated token, which the code of Yang et al. (2026) reads; the first-token score is kept beside it.

4. *Control*: the same patch at the same heads, with the mean output of a different task. A head set carries the task if its own mean raises accuracy and the control mean does not; a gain under both points to the answer format rather than the task.

Three head sets are patched for each task and style: $H^{\mathrm{L}}_{t,s}$, $H^{\mathrm{J}}_{t,s}$, and the heads found only by J-head, $H^{\mathrm{J}}_{t,s} \setminus H^{\mathrm{L}}_{t,s}$. We report the accuracy at each α averaged over the wordings of the mean's style. For every task the patch is applied at the last prompt position only, as in Yang et al. (2026); the reply is then generated without it, and its answer is scored. Which items are patched and which means and control tasks are used are given in E1 (Sec. 5.2).

4.5 Answer evaluation

We score every reply in more than one way because of what Hua et al. (2025) show about scoring and prompt wording. They evaluate seven LLMs, Llama-3.1-8B-Instruct among them, on six benchmarks under twelve prompt templates each. Each reply is scored twice: once with the benchmark's usual rule (log-likelihood over the answer options, or extraction and matching of the answer string), and once by an LLM judge that sees the question, the reference answer and

the reply. Under the rules, accuracy varies much more across templates than under the judge (Gemma-2 on ARC-Challenge: standard deviation 0.28 against 0.005). The ranking of the models also changes from template to template (mean Spearman correlation between templates 0.30 against 0.92). The rules mark answers wrong that are correct but phrased or formatted differently. Much of the measured sensitivity to prompt wording therefore comes from the scoring, not from the model. A rule built to accept equivalent answers (symbolic equivalence on MATH) varies as little as the judge. The judge's verdicts agree with those of three human annotators (Cohen's κ 0.92 over all benchmarks, lowest 0.67 on NarrativeQA) and with those of a second judge model. This matters here because E0 compares prompt styles and wordings by accuracy, and a difference between wordings may only be a difference in how the answers are written. The source evaluation we use is a rigid matcher: Yang et al. (2026) compare only the first generated token with the first token of the single reference answer. A capitalized German noun (*Gemeinde* for the reference *gemeinde*), an answer in quotes, or another valid synonym is therefore wrong, and a first token that only begins the reference (*Niger* for *Nigeria*) is right. For a numeric reference the first token is the leading space alone (the tokenizer splits `" 2"` into `" "` and `"2"`), so any number counts as correct: in E0 all 29 numeric items of next-item and of prev-item are scored correct under every instruction template. In a first check on the stored E0 replies, most of the spread across instruction templates on the translation and next-item tasks goes once case, quotes and leading articles are ignored. We therefore evaluate every E0 prompt in two ways, from one saved output, and also with a flexible score that ignores how the answer is written.

1. *Saved output.* Every reply is generated once, with greedy decoding, and saved with its prompt, input, reference answer, generated token ids and text, number of tokens, and why generation stopped, so that every score and analysis below can be recomputed without running the model again. Replies are generated from raw text, without the chat template, in batches of 20 as in Yang et al. (2026), with the padding masked. A reply ends at the first of: the delimiter `;` of the example prompts, a newline once the reply has begun, the end-of-sequence token, or 32 new tokens, twice the longest reference answer (15 tokens). Under greedy decoding the first tokens of a reply do not depend on how long generation runs, so the reply cut to a shorter limit is the reply that limit would give.

2. *Strict score*: the source evaluation, unchanged, on the full reply: the first generated token against the first token of the reference (Yang et al., 2026). Computed on the replies cut to the E0 limits, it must reproduce the E0 accuracies; differences can only come from batch numerics and are reported.

3. *Flexible score*: the reply, up to where it stops, begins with the whole reference answer, as words. Case, accents, punctuation and one leading article or *to* are ignored. A hyphen or apostrophe inside a word joins it (*non-corresponding* matches *noncorresponding*), a possessive *'s* is dropped, and any other punctuation separates words (*petit(e)* matches *petit*); a part of the reference in parentheses is optional (*Euro* matches *Euro (EUR)*). It removes the formatting errors of the first-token score (case, quotes, articles) and its false positives (*Niger* for *Nigeria*). Another spelling of a name (*Luxembourg* for *Luxembourg City*) and another valid answer remain wrong; these are left to the judge.

4. *Judged score*: an LLM judge (GPT-5.5) receives a neutral description of the task (not the prompt wording being evaluated), the item's input, the reference answer and the full reply, but not the strict score. It returns whether the reply's final answer is correct, the answer it extracted, and, where it disagrees with the strict score, the reasons (below). The judge evaluates the final answer, not the reasoning, accepts a different answer when the task admits one (another synonym, antonym or translation), and counts two conflicting final answers as wrong. Every reply is judged, including those the strict score accepts.

5. *Straight away*: the reply begins with the answer the judge extracted, after leading whitespace, rather than with a quote, an article or another preamble.

Reasons for disagreement. Each reply that the strict score marks wrong and the judge right gets one or more reasons from three groups; its primary reason is the first that applies, in the order below. Reasons that a rule can detect (a missing box, case, punctuation) are assigned by rule; the judge assigns the others.

1. *Where the answer is*: the strict score reads a part of the reply that does not hold the final answer.
    - *Leading text*: text before the answer (*The answer is*, an article, the input repeated) where the score reads the first token or word.
    - *Earlier answer*: the score read an earlier answer, which the model then revised or restated correctly.
2. *How the answer is written*: the score finds the answer but does not accept its form.
    - *Case*: the answer differs from the reference only in capitalization.
    - *Extra characters*: quotes, a final period, or Markdown or LaTeX around the answer.
    - *Tokenization*: the same word, whose first token differs from the first token of the reference.
    - *Surface variant*: the same answer written differently: diacritics, number format or spelled-out numbers, inflection, abbreviation.
3. *Which answer is right*: a *valid alternative*, a different answer that is also correct, which a single reference cannot accept; or a *reference error*, a reference that appears wrong, checked by hand.
4. *Other*.

Replies that the strict score marks right and the judge wrong get reasons from the same groups: a *first-token false positive* (a first token that only begins the reference, as *Niger* for *Nigeria*) is a matter of how the answer is written.

What changes. Head identification, $P^{+}_{t,s}$ and the causal checks use the flexible score, so the selection of the identification wordings, the head sets, the means, the patched items and the scoring of the patched replies are redone together. The judged score is not used for them: it also accepts other valid answers, which the reference of the patched items does not contain. Under the flexible score the most accurate instruction template changes on antonym (template 3 to 2), country–capital (1 to 3) and English–German (3 to 1).

Second check. Yang et al. (2026, Sec. 2.4.1, App. H.1.3, Figs. 7 and 67) also patch prompts that give the task and are answered wrongly, whose unpatched accuracy is 0, with the mean output of prompts answered correctly. Their paper does not say which wrongly answered prompts are patched. Their one released script for the causal checks (`causal_fix_incorrect.py`) patches the 1-shot prompts of the items answered wrongly with one shot, and we follow it. These prompts show the answer format. Head sets, means, α and scoring are as above, and the script has no control. The figures that App. G.1 attributes to patching zero-shot prompts (Figs. 57 and 62) reproduce only in this setting (Sec. 5.2.2). We report both checks.

4.6 Measures

For each task t, style s and readout R:

- *Accuracy* of each prompt, with the flexible score (Sec. 4.5); in E0 also with the strict score, the evaluation of Yang et al. (2026), and the judged score.
- *Head count* $|H^R_{t,s}|$ and the distribution of the heads over layers.
- *Overlap* between two head sets A and B: $|A \cap B|/|A|$ and $|A \cap B|/|B|$, and the overlap ratio of the code of Yang et al. (2026), whose paper does not state it: $|A \cap B| / \min(|A|, |B|)$. Compared pairs: $H^{\mathrm{L}}_{t,s}$ and $H^{\mathrm{J}}_{t,s}$ (RQ1); $H^R_{t,\mathrm{rule}}$ and $H^R_{t,\mathrm{example}}$.
- *Causal effect*: the highest accuracy over α minus the unpatched accuracy, for the head set and for the control.
- *Copy share*: the share of prompts on which a head's matched terms $(T^R_{k,l,h}(p) \cap V_t)$ occur in the prompt text, averaged over the heads of $H^R_{t,s}$. It separates heads that may copy task words from the prompt from heads that do not.

Experiments and reported results. Every experiment is run once, with greedy decoding, on Llama-3.1-8B-Instruct. Each is described in its own section by the conditions it crosses and the steps it runs, followed by its results. The tables and figures are generated from the result files by `scripts/results_tex.py`. E0 accuracies are the strict scores of the saved outputs (Sec. 4.5). Head sets and causal checks use the flexible score (Sec. 4.5, "What changes"): P^+, the means and the patched items are chosen by it, and causal-check accuracies are the flexible scores of the replies generated from the patched prompts, over the patched items.

5 Replication of Yang et al. (2026) under both readouts

RQ1 How do the per-head logit lens and Head-JLens compare as readouts for identifying lexical task heads on the simple tasks?

Hypothesis 1 (replication)**.** *On the simple tasks, Logit-head reproduces Yang et al. (2026): lexical task heads are found for every task, and patching them into prompts without task information raises accuracy above the unpatched baseline.* Falsified *if, on the same model, the method finds no heads or no accuracy gain for tasks where Yang et al. (2026) report both.*

Hypothesis 2 (readouts)**.** *Following Merlo (2026), on the simple tasks J-head identifies at least the lexical task heads that Logit-head identifies, and the heads found only by J-head have a causal effect above baseline.* Falsified *if the J-head set misses heads of the Logit-head set, or if the heads found only by J-head have no causal effect.*

5.1 E0: Behavioural baseline

5.1.1 Design

Question. On which tasks and prompt styles does the model perform above chance, and does it answer straight away?

Conditions. The 14 tasks, each on every item of its dataset, under eleven prompts: the five instruction templates (rule style), 1, 5, 10, 20 and 30 shots (example style), and the zero-shot prompt (no task information).

Procedure.

1. Generate the model's answer to every prompt and score it with the evaluation of the source paper: the first generated token against the first token of the correct answer.
2. Compare each accuracy with chance, 0 for these open answers.
3. Evaluate every prompt again from its saved output (Sec. 4.5): the strict score, the flexible score, the judged score, and the reasons for each disagreement.

Output. Accuracy per task and prompt (Table 1); for every task and prompt, the strict and judged accuracy, the share answered straight away, the four counts of agreement and disagreement between the two scores, and the disagreements by reason; for every task, the standard deviation of accuracy across the prompts of a style under each score (Hua et al., 2025), the Spearman correlation between the two scores' rankings of these prompts, and whether the most accurate instruction template changes; for each task and style, the prompts $P^{+}_{t,s}$ used by E1 and, for the rule style, the most accurate template from which they are taken; the items answered correctly and wrongly, which select the patched items of the causal checks.

5.1.2 Results

Saved outputs. The strict score of the saved outputs reproduces the E0 accuracies item by item: it agrees with the first E0 run on every item of the 154 prompts. The judged score has not been run, so the outputs that depend on it (judged accuracy, the share answered straight away, the disagreements and their reasons, and the comparison of scores across prompts after Hua et al., 2025) are not reported yet.

Simple tasks (Table 1). Every task is answered well above its zero-shot accuracy under both styles, and the zero-shot prompts are almost never answered correctly. Accuracy rises from one to five shots and changes little after that. The instruction templates differ most on the translation tasks, next-item, prev-item, park–country and synonym, where one or two templates fall well below the others.

Under the flexible score (Table 2), most of the spread across instruction templates on the three translation tasks goes away: their weakest templates answer in quotes or with capitalized German nouns. On next-item, prev-item and park–country the spread stays, so there the weak templates are weaker in what they answer, not only in how. German example prompts gain at every number of shots, and antonym and synonym gain on their weakest template; the other tasks change little. On numeric next-item items, which the first-token score accepts whatever the number, three of the five templates lose between 7 and 14 of 29 items.

		Rule (5 templates)		Example (shots)					
Task	Items	mean	range	1	5	10	20	30	Zero-shot
antonym	2398	0.62	0.53–0.65	0.32	0.66	0.69	0.69	0.69	0.00
synonym	2880	0.44	0.29–0.51	0.26	0.44	0.47	0.51	0.52	0.00
singular–plural	205	0.98	0.96–0.99	0.89	0.98	0.99	0.99	0.99	0.00
present–past	293	0.95	0.84–0.99	0.79	0.98	0.98	0.99	0.99	0.00
country–capital	197	0.98	0.96–0.98	0.87	0.98	0.97	0.97	0.97	0.00
country–currency	197	0.74	0.68–0.82	0.69	0.74	0.79	0.81	0.84	0.01
landmark–country	836	0.84	0.77–0.88	0.77	0.90	0.90	0.91	0.91	0.00
park–country	749	0.79	0.48–0.91	0.94	0.97	0.97	0.99	0.99	0.00
product–company	522	0.73	0.64–0.81	0.70	0.84	0.85	0.87	0.89	0.04
next-item	225	0.42	0.20–0.59	0.56	0.98	0.97	0.98	0.99	0.00
prev-item	225	0.54	0.20–0.74	0.54	0.96	0.96	0.96	0.97	0.00
English–French	4698	0.62	0.19–0.79	0.60	0.81	0.82	0.82	0.83	0.00
English–German	5145	0.41	0.10–0.51	0.39	0.53	0.53	0.56	0.56	0.00
English–Spanish	5199	0.71	0.25–0.84	0.69	0.84	0.84	0.85	0.85	0.00

Table 1: E0, simple tasks: strict accuracy per prompt style. Rule: the five instruction templates of Yang et al. (2026), mean and range; example: by number of shots; zero-shot: the prompts without task information. Items: dataset size.

		Rule (5 templates)		Example (shots)					
Task	Items	mean	range	1	5	10	20	30	Zero-shot
antonym	2398	0.66	0.63–0.68	0.32	0.66	0.68	0.69	0.69	0.00
synonym	2880	0.45	0.39–0.50	0.27	0.43	0.47	0.50	0.51	0.01
singular–plural	205	0.98	0.97–0.99	0.87	0.98	0.98	0.99	0.99	0.00
present–past	293	0.96	0.89–0.99	0.79	0.98	0.98	0.99	0.99	0.00
country–capital	197	0.97	0.95–0.98	0.86	0.96	0.96	0.96	0.96	0.00
country–currency	197	0.72	0.68–0.78	0.67	0.71	0.77	0.79	0.81	0.00
landmark–country	836	0.85	0.77–0.88	0.77	0.90	0.90	0.91	0.91	0.00
park–country	749	0.79	0.48–0.89	0.94	0.97	0.97	0.99	0.99	0.00
product–company	522	0.73	0.64–0.81	0.70	0.84	0.85	0.87	0.89	0.05
next-item	225	0.42	0.21–0.66	0.54	0.98	0.97	0.98	0.99	0.00
prev-item	225	0.53	0.20–0.74	0.46	0.96	0.96	0.96	0.97	0.00
English–French	4698	0.74	0.71–0.78	0.58	0.80	0.81	0.81	0.81	0.00
English–German	5145	0.70	0.61–0.75	0.52	0.72	0.72	0.73	0.73	0.00
English–Spanish	5199	0.79	0.71–0.84	0.67	0.83	0.83	0.84	0.84	0.00

Table 2: E0, simple tasks: flexible accuracy per prompt style (Sec. 4.5), on the same saved replies and with the same layout as Table 1.

5.2 E1: Lexical task heads under both readouts

5.2.1 Design

Question. H1, RQ1 and H2: does Logit-head reproduce Yang et al. (2026) on our setup, and how does J-head differ from it where the method is known to work?

Conditions. 14 tasks × 2 styles (rule, example) × 2 readouts (Logit-head, J-head). Each pair of task and style gives three head sets: $H^{\mathrm{L}}_{t,s}$, $H^{\mathrm{J}}_{t,s}$, and the heads found only by J-head, $H^{\mathrm{J}}_{t,s} \setminus H^{\mathrm{L}}_{t,s}$.

Procedure.

1. *Identification prompts*: $P^{+}_{t,\mathrm{rule}}$, the first 100 prompts answered correctly under the most accurate of the five instruction templates; $P^{+}_{t,\mathrm{example}}$, the first 100 answered correctly with

five shots.

2. *Head sets*: decode every head at the last prompt token under both readouts and apply Definition 1.

3. *Measures* (Sec. 4.6): head count and layers of each set; overlap of the Logit-head and J-head sets of each style (RQ1), and of the rule-style and example-style sets of each readout; copy share.

4. *First causal check* (Sec. 4.4). Patched items: the first 100 items answered correctly with one shot. Patched prompts: their zero-shot prompts, whose unpatched accuracy is the baseline. Means: one per source wording, each over the first 100 prompts of that wording answered correctly, leaving out the patched items; there are ten, the five instruction templates and the five shot counts. Each head set is patched with the five means of its own style and, across styles, with the five means of the other style. Control: the mean of the control task of Yang et al. (2026) at the same heads (their code maps country–capital, product–company and country–currency to park–country, park–country to country–currency, landmark–country to product–company, English–Spanish to English–German, and pairs antonym with synonym, singular–plural with present–past, next-item with prev-item and English–French with English–German), one per style of mean, taken as in their code: over the control task's 1-shot prompts, at all the items it answers correctly with one shot (for example-style means) or under the second instruction template (for rule-style means). Scaling: the 15 values of α in Sec. 4.4; $\alpha = 1$ writes the mean itself, $\alpha = 0$ sets the heads to zero, and a negative α reverses its sign.

5. *Second causal check* (Sec. 4.4). Patched items: the first 100 items answered wrongly with one shot, or all of them where fewer are wrong. Patched prompts: their 1-shot prompts, whose unpatched accuracy is 0 by construction. Head sets, means and α as in the first check; no control.

6. *Scoring*: the reply generated from each patched prompt, as the saved outputs of the simple tasks (Sec. 4.5), scored by the flexible score; the first-token score is kept beside it. Accuracy at each α is averaged over the five means of a style; the causal effect is the highest of these averages minus the baseline.

Output. Head sets and measures (Tables 3 and 4, Figures 1 and 2); causal effects with means of the same style (Tables 5 and 7) and of the other style (Tables 6 and 8); accuracy against α (Figures 3 and 4).

5.2.2 Results

Head sets (Table 3, Figures 1 and 2). Logit-head finds lexical task heads for every task under both styles, from two heads (synonym) upward. These sets are smaller than those that Yang et al. (2026) report for the same model in their Figs. 28 (instruction-based prompts) and 20 (example-based) on almost every task (Table 4). Their paper gives other counts for the same model in Figs. 79 and 88, which name no prompt style. Run in the setting of their code (attended padding and the first-token score, Sec. 4.3), our rule-style counts are within one head of Fig. 79 on every task, and our example-style counts within one head of Fig. 88. E1 differs from that run only in the padding mask and the flexible score, which change the most accurate instruction template on eight and on three tasks and change the prompts read on all of them. J-head finds more heads than Logit-head on 23 of the 28 pairs of task and style. Nearly two thirds of the heads it adds lie in layers 10 to 19, whereas most Logit-head heads lie in layer 20 or later (Figure 1). The J-head set contains the whole Logit-head set on 4 of the 28 pairs, and at least half of it on every pair. Rule-style sets are at least as large as example-style sets on every task except

country–capital under J-head, where they differ by one head, and most heads of an example-style set are also in the rule-style set, except on synonym under both readouts and singular–plural under Logit-head. With the overlap ratio of the code of Yang et al. (2026), the shared heads over the smaller of the two sets, rule-style and example-style sets share 70% of their heads under Logit-head and 69% under J-head, averaged over the 14 tasks; Yang et al. (2026, Fig. 3) report 73% over 17 tasks. As a share of the rule-style set the averages are 43% and 45%, and as a share of the example-style set 70% and 69%. The copy share is high under most rule-style prompts, whose instructions contain task words, and lower on antonym, country–capital, singular–plural and synonym, where the heads also decode task words that the instruction does not contain; under example-style prompts it is close to zero except on country–capital (Figure 2).

Causal check (Tables 5 and 6, Figure 3). Patching the mean output of the Logit-head set, with means of its own style, raises accuracy above the unpatched baseline by at least ten items on country–capital, landmark–country, park–country and product–company under both styles, and on country–currency, next-item and singular–plural with examples. It has almost no effect on antonym, synonym, present–past, prev-item and the three translation tasks. Gains appear at positive α; averaged over tasks and head sets, accuracy is highest at $\alpha = 4$. The control mean raises accuracy much less than the task mean, except on park–country, where it raises it by a comparable amount, and on next-item, where at the J-head set and at the heads found only by J-head it raises it as much or more. The control task of park–country is country–currency and that of next-item is prev-item; neither shares a term with the term list of the task.

The J-head set raises accuracy as much as the Logit-head set or more on most tasks, and also on synonym and prev-item with examples, where the Logit-head set does not; it raises it less on landmark–country with examples and on country–capital with rules. The heads found only by J-head raise accuracy by themselves on most tasks where the J-head set does, most strongly on singular–plural and country–capital with examples and on landmark–country with rules. The control mean at those heads raises it less, except on next-item.

Patched with the means of the other style, the head sets raise accuracy on the same tasks, often by a comparable amount or more. On singular–plural, the rule-style J-head set and the heads found only by J-head raise accuracy with example-style means (to 0.74 and 0.82) but hardly with their own; the rule-style Logit-head set does neither. Antonym, present–past and the translation tasks gain at most six items under any pairing.

Second causal check (Tables 7 and 8, Figure 4). Every head set raises accuracy above zero on every task, under both styles and with means of either style, from a few items on synonym to most of the patched items on singular–plural, country–capital and, with the example-style J-head sets, next-item and prev-item. This includes antonym, present–past and the translation tasks, where patching the zero-shot prompts has almost no effect. Under Logit-head the gains are largest on singular–plural, country–capital and present–past and smallest on synonym and the translation tasks, the order reported by Yang et al. (2026, Figs. 57 and 62), and as there park–country also gains at negative α. On landmark–country, park–country and product–company the gains are smaller than with the zero-shot prompts. The J-head set raises accuracy about as much as the Logit-head set or more on most tasks, and far more on next-item and prev-item with examples; it raises it less on present–past with rules. The heads found only by J-head raise it on every task where that set is not empty, least on landmark–country with examples.

Hypotheses. For H1, lexical task heads are found for every simple task. With the zero-shot prompts of Yang et al. (2026, App. G.1), patching the Logit-head set raises accuracy by at least ten items on seven tasks with means of its own style; there is no such gain on antonym, synonym, present–past, prev-item or the three translation tasks, and on park–country the control mean raises accuracy as well. With the wrongly answered prompts of their Sec. 2.4.1 and App. H.1.3,

taken as 1-shot prompts as in their released script, patching raises accuracy on all fourteen tasks, in the per-task pattern of their Figs. 57 and 62. H1 is therefore falsified for the zero-shot patching of App. G.1, and the figures shown there reproduce only in the setting of the second check: the gains they show on present–past and the translation tasks need prompts that give the task and show the answer format. Scored by the first generated token instead, the verdict is the same: the same seven tasks gain by at least ten items in the first check, and all fourteen gain in the second. For H2, the first condition fails, since the J-head set misses part of the Logit-head set on 24 of 28 pairs, and the second holds in both checks: heads found only by J-head have causal effects of their own. H2 is therefore falsified by its first condition.

	Rule			Example			Rule ∩ example	
Task	L	J	L∩J	L	J	L∩J	L	J
antonym	14	21	12	10	15	9	7	12
synonym	2	22	1	2	11	1	0	5
singular–plural	20	24	14	19	20	12	8	12
present–past	10	7	6	6	4	4	4	3
country–capital	21	35	19	13	36	13	12	23
country–currency	17	33	16	15	27	13	13	21
landmark–country	23	27	19	16	14	10	12	9
park–country	24	24	18	14	12	9	12	9
product–company	30	40	25	16	25	12	12	16
next-item	35	47	29	9	17	6	6	13
prev-item	20	35	19	7	17	5	4	9
English–French	11	22	9	5	15	5	4	11
English–German	7	19	6	6	14	5	5	13
English–Spanish	6	14	6	3	10	3	3	6

Table 3: E1: lexical task heads per task and style under Logit-head (L) and J-head (J). L∩J: heads in both sets. Rule ∩ example: heads in both the rule-style and the example-style set of a readout. Identification prompts: 100 per task and style, the first answered correctly under the most accurate instruction template (rule) and under 5-shot prompts (example).

	Rule				Example			
Task	Fig. 28	Fig. 79	Source	E1	Fig. 20	Fig. 88	Source	E1
antonym	23	17	17	14	12	9	9	10
synonym	7	2	2	2	8	4	4	2
singular–plural	26	18	18	20	24	15	15	19
present–past	11	9	9	10	6	6	5	6
country–capital	31	27	28	21	15	13	13	13
country–currency	20	16	17	17	16	15	15	15
landmark–country	25	21	21	23	14	12	12	16
park–country	26	22	22	24	16	9	9	14
product–company	35	28	28	30	20	16	15	16
next-item	42	35	36	35	16	11	11	9
prev-item	29	25	25	20	10	6	5	7
English–French	14	11	11	11	6	5	5	5
English–German	11	9	9	7	7	6	6	6
English–Spanish	9	7	7	6	3	3	3	3
Sum	309	247	250	240	173	130	127	141

Table 4: E1: Logit-head counts against those of Yang et al. (2026) for Llama-3.1-8B-Instruct, read off the bar labels of their Figs. 28 (instruction-based prompts) and 20 (example-based), the Instruct bars of Fig. 79 and the bars for the original term lists of Fig. 88; Figs. 79 and 88 name no style and are placed by their match. Source: our sets in the setting of their code, with attended padding and the first-token score. E1: our sets (Table 3).

			Task mean			Control mean		
Task	Style	Baseline	L	J	J only	L	J	J only
antonym	rule	0.00	0.04	0.03	0.01	0.00	0.00	0.00
	example		0.06	0.04	0.01	0.00	0.00	0.00
synonym	rule	0.03	0.01	0.12	0.13	0.01	0.02	0.03
	example		0.02	0.33	0.33	0.03	0.09	0.07
singular–plural	rule	0.00	0.03	0.02	0.01	0.00	0.03	0.02
	example		0.33	0.90	0.89	0.05	0.05	0.04
present–past	rule	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	example		0.00	0.00	–	0.00	0.00	–
country–capital	rule	0.00	0.70	0.59	0.05	0.01	0.00	0.00
	example		0.72	0.98	0.72	0.02	0.01	0.03
country–currency	rule	0.00	0.06	0.12	0.07	0.00	0.00	0.00
	example		0.10	0.23	0.17	0.00	0.00	0.00
landmark–country	rule	0.00	0.79	0.75	0.39	0.00	0.09	0.12
	example		0.74	0.42	0.01	0.00	0.00	0.04
park–country	rule	0.00	0.39	0.48	0.19	0.29	0.32	0.14
	example		0.59	0.54	0.00	0.34	0.20	0.00
product–company	rule	0.11	0.24	0.35	0.12	0.01	0.00	0.04
	example		0.23	0.28	0.21	0.01	0.06	0.06
next-item	rule	0.00	0.01	0.17	0.27	0.00	0.35	0.50
	example		0.29	0.60	0.54	0.00	0.45	0.64
prev-item	rule	0.00	0.04	0.04	0.05	0.00	0.00	0.01
	example		0.00	0.20	0.25	0.00	0.01	0.01
English–French	rule	0.00	0.00	0.02	0.02	0.01	0.00	0.01
	example		0.00	0.01	0.01	0.01	0.02	0.03
English–German	rule	0.00	0.00	0.01	0.04	0.00	0.00	0.00
	example		0.00	0.00	0.00	0.00	0.00	0.00
English–Spanish	rule	0.00	0.00	0.01	0.01	0.00	0.00	0.00
	example		0.00	0.01	0.01	0.00	0.00	0.00

Table 5: E1 causal check: causal effect, the highest accuracy over $\alpha \in [-10, 10]$ minus the unpatched accuracy (baseline), on the 100 patched items of each task; accuracy is the flexible score of the reply generated from each patched prompt. The head set of each style is patched with means of the same style, and accuracy is averaged over the style's five source wordings before the maximum is taken. Task mean: the head set's mean output over the task's own prompts; control mean: over the control task's prompts (Sec. 5.2). L: Logit-head set; J: J-head set; J only: heads in the J-head set and not in the Logit-head set. –: empty set.

			Task mean			Control mean		
Task	Heads	Baseline	L	J	J only	L	J	J only
antonym	rule	0.00	0.06	0.06	0.00	0.00	0.00	0.00
	example		0.03	0.03	0.01	0.00	0.00	0.00
synonym	rule	0.03	0.01	0.05	0.06	0.01	0.01	0.03
	example		0.02	0.18	0.17	0.03	0.08	0.08
singular–plural	rule	0.00	0.02	0.74	0.82	0.01	0.03	0.02
	example		0.09	0.05	0.01	0.04	0.06	0.04
present–past	rule	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	example		0.00	0.00	–	0.00	0.00	–
country–capital	rule	0.00	0.61	0.65	0.06	0.01	0.00	0.00
	example		0.77	0.89	0.66	0.02	0.01	0.03
country–currency	rule	0.00	0.09	0.24	0.21	0.00	0.00	0.00
	example		0.06	0.14	0.07	0.00	0.00	0.00
landmark–country	rule	0.00	0.70	0.97	0.95	0.00	0.12	0.13
	example		0.89	0.86	0.02	0.00	0.00	0.05
park–country	rule	0.00	0.33	0.64	0.15	0.30	0.32	0.15
	example		0.52	0.57	0.01	0.33	0.21	0.00
product–company	rule	0.11	0.33	0.37	0.12	0.01	0.01	0.04
	example		0.25	0.31	0.21	0.01	0.05	0.06
next-item	rule	0.00	0.26	0.52	0.43	0.00	0.29	0.44
	example		0.00	0.30	0.30	0.00	0.53	0.62
prev-item	rule	0.00	0.00	0.01	0.01	0.00	0.00	0.01
	example		0.00	0.12	0.10	0.00	0.00	0.02
English–French	rule	0.00	0.00	0.01	0.01	0.00	0.00	0.01
	example		0.00	0.00	0.01	0.01	0.01	0.04
English–German	rule	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	example		0.00	0.00	0.00	0.00	0.00	0.00
English–Spanish	rule	0.00	0.00	0.01	0.01	0.00	0.00	0.00
	example		0.00	0.01	0.01	0.00	0.00	0.01

Table 6: E1 causal check across styles, as Table 5, with the head set of each style (Heads) patched with the means of the other style: rule-style heads with the means of the five shot counts, example-style heads with the means of the five instruction templates.

			Task mean			Control mean		
Task	Style	Baseline	L	J	J only	L	J	J only
antonym	rule	0.00	0.33	0.34	0.18	–	–	–
	example		0.33	0.35	0.24	–	–	–
synonym	rule	0.00	0.04	0.12	0.14	–	–	–
	example		0.03	0.12	0.12	–	–	–
singular–plural	rule	0.00	0.56	0.72	0.46	–	–	–
	example		0.77	0.87	0.85	–	–	–
present–past	rule	0.00	0.43	0.35	0.18	–	–	–
	example		0.61	0.57	–	–	–	–
country–capital	rule	0.00	0.77	0.79	0.62	–	–	–
	example		0.71	0.72	0.65	–	–	–
country–currency	rule	0.00	0.21	0.17	0.19	–	–	–
	example		0.31	0.29	0.25	–	–	–
landmark–country	rule	0.00	0.34	0.32	0.32	–	–	–
	example		0.34	0.31	0.05	–	–	–
park–country	rule	0.00	0.24	0.24	0.20	–	–	–
	example		0.43	0.41	0.26	–	–	–
product–company	rule	0.00	0.14	0.22	0.25	–	–	–
	example		0.17	0.37	0.31	–	–	–
next-item	rule	0.00	0.23	0.45	0.42	–	–	–
	example		0.43	0.81	0.81	–	–	–
prev-item	rule	0.00	0.11	0.14	0.16	–	–	–
	example		0.15	0.80	0.80	–	–	–
English–French	rule	0.00	0.10	0.20	0.20	–	–	–
	example		0.12	0.17	0.19	–	–	–
English–German	rule	0.00	0.09	0.18	0.20	–	–	–
	example		0.13	0.13	0.21	–	–	–
English–Spanish	rule	0.00	0.10	0.27	0.24	–	–	–
	example		0.04	0.15	0.15	–	–	–

Table 7: E1, second causal check, as Table 5, with the 1-shot prompts of the first 100 items answered wrongly with one shot patched; fewer items are wrong on singular–plural (26), country–capital (27), park–country (45), present–past (61) and country–currency (65). Unpatched accuracy is 0 by construction. No control mean.

			Task mean			Control mean		
Task	Heads	Baseline	L	J	J only	L	J	J only
antonym	rule	0.00	0.32	0.38	0.16	–	–	–
	example		0.25	0.28	0.22	–	–	–
synonym	rule	0.00	0.04	0.09	0.10	–	–	–
	example		0.03	0.18	0.18	–	–	–
singular–plural	rule	0.00	0.65	0.83	0.82	–	–	–
	example		0.68	0.75	0.29	–	–	–
present–past	rule	0.00	0.56	0.41	0.22	–	–	–
	example		0.45	0.42	–	–	–	–
country–capital	rule	0.00	0.72	0.71	0.42	–	–	–
	example		0.79	0.82	0.73	–	–	–
country–currency	rule	0.00	0.30	0.26	0.18	–	–	–
	example		0.22	0.24	0.19	–	–	–
landmark–country	rule	0.00	0.32	0.38	0.38	–	–	–
	example		0.32	0.32	0.08	–	–	–
park–country	rule	0.00	0.31	0.22	0.19	–	–	–
	example		0.34	0.36	0.21	–	–	–
product–company	rule	0.00	0.25	0.32	0.30	–	–	–
	example		0.17	0.31	0.32	–	–	–
next-item	rule	0.00	0.54	0.72	0.72	–	–	–
	example		0.08	0.47	0.45	–	–	–
prev-item	rule	0.00	0.36	0.26	0.15	–	–	–
	example		0.07	0.29	0.39	–	–	–
English–French	rule	0.00	0.10	0.19	0.20	–	–	–
	example		0.10	0.16	0.17	–	–	–
English–German	rule	0.00	0.13	0.15	0.21	–	–	–
	example		0.09	0.13	0.17	–	–	–
English–Spanish	rule	0.00	0.05	0.21	0.19	–	–	–
	example		0.09	0.17	0.17	–	–	–

Table 8: E1, second causal check across styles, as Table 6.

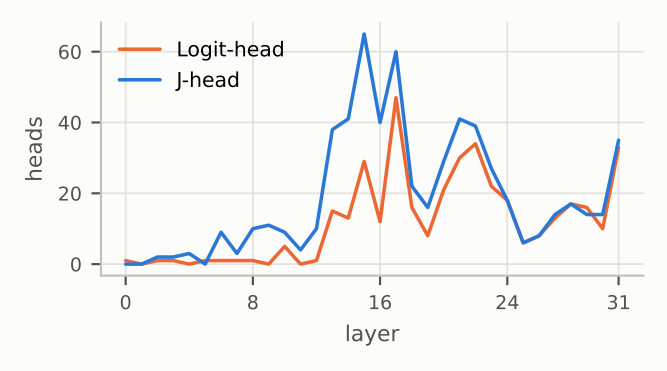


Figure 1: Lexical task heads per layer under each readout, summed over the head sets of all simple tasks and styles, at the last prompt token.

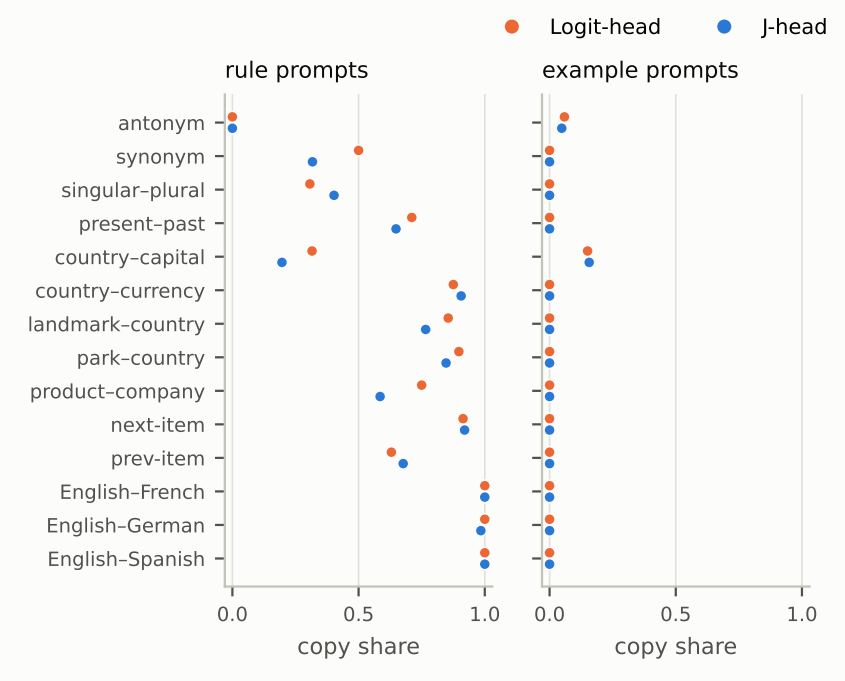


Figure 2: Copy share of each simple-task head set (Sec. 4.6): the share of prompts on which a head's matched terms occur in the prompt text, averaged over the set.

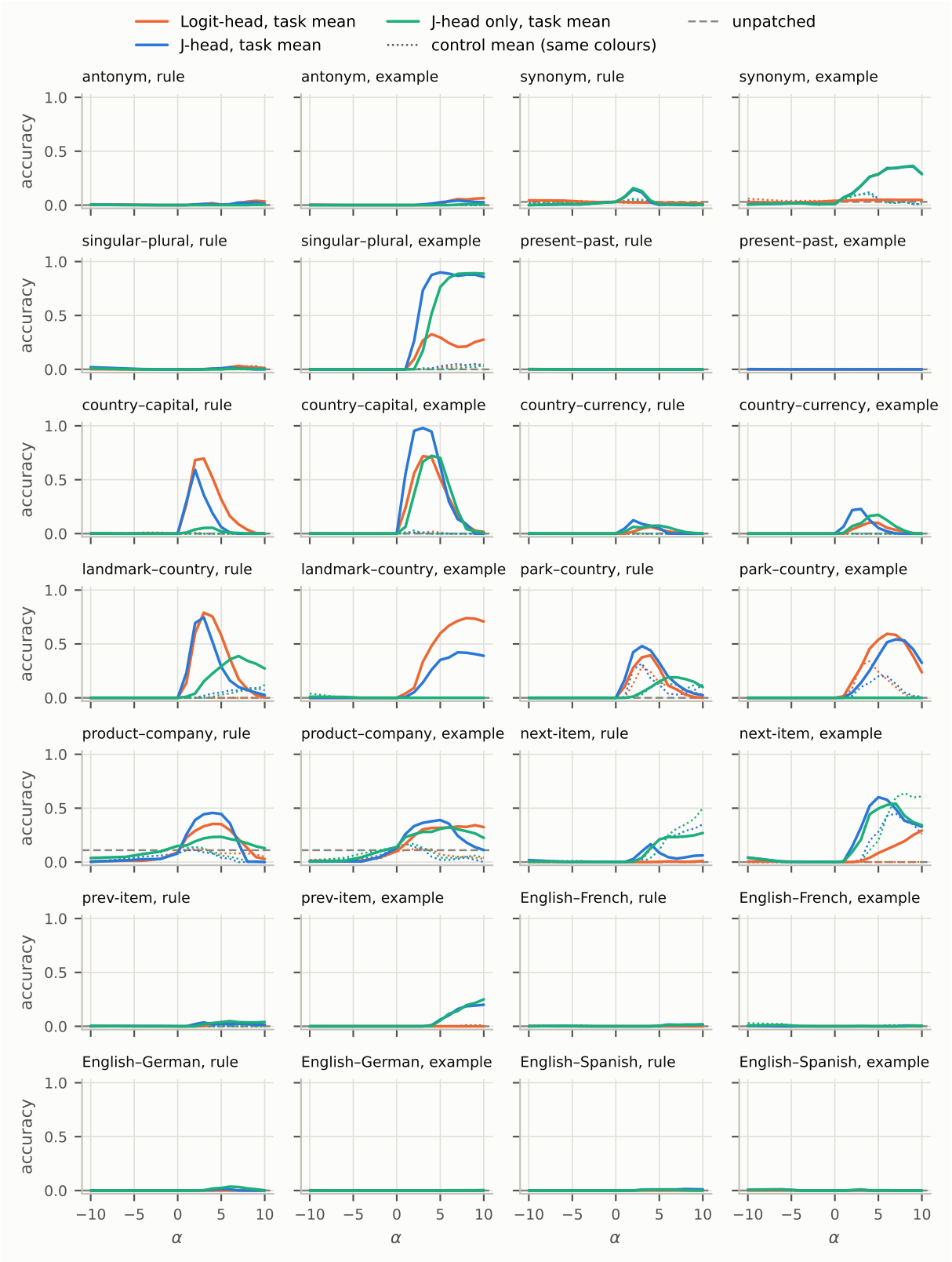


Figure 3: E1 causal check: accuracy on the 100 patched items of each task against α, with each style's head set patched with means of the same style, averaged over the style's five source wordings. Solid: the head set's mean over the task's prompts; dotted: the control mean at the same heads; dashed: unpatched accuracy.

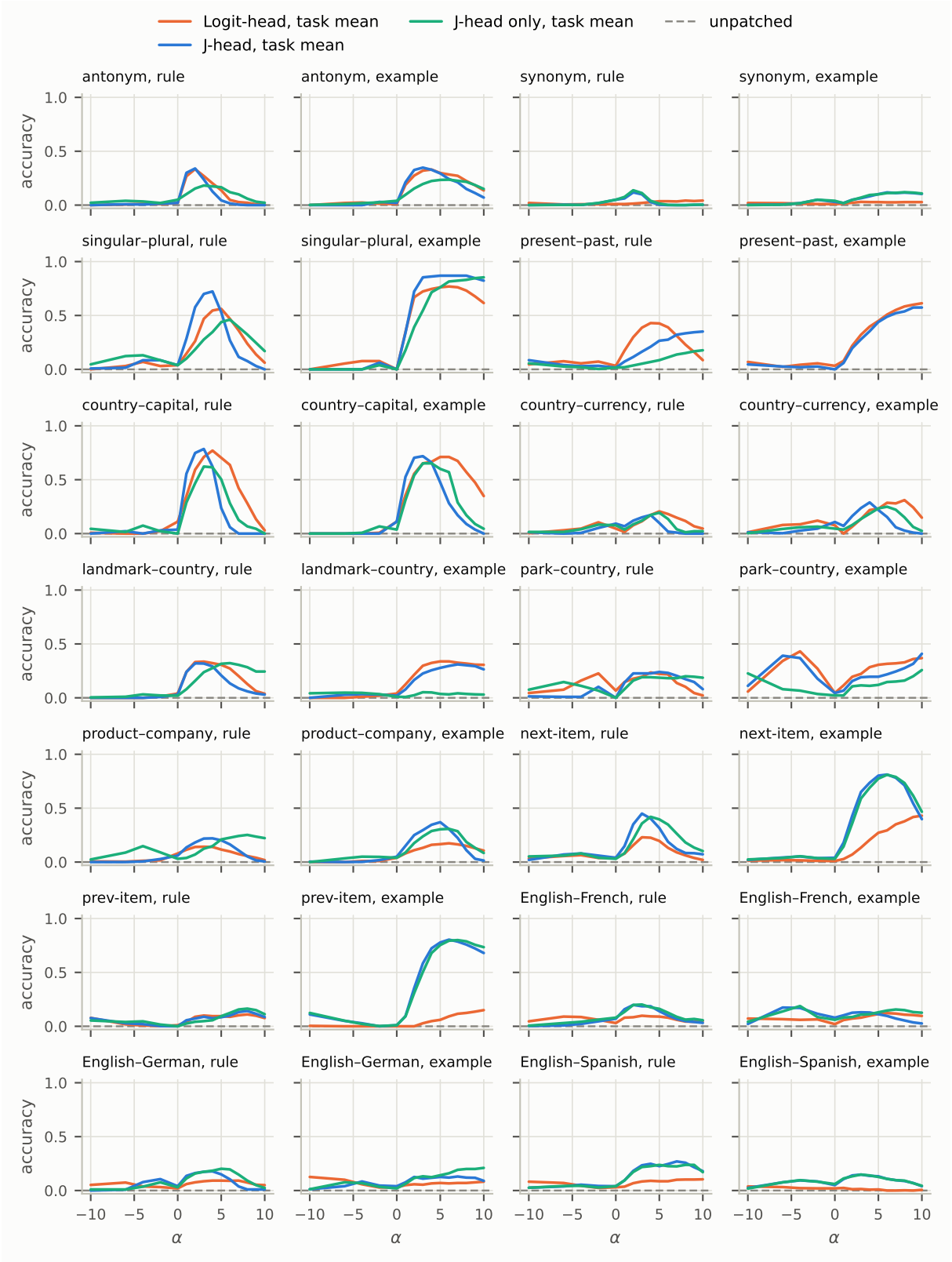


Figure 4: E1, second causal check, as Figure 3, with the wrongly answered 1-shot prompts patched. No control mean.

6 E1b: Heads from all five instruction templates

Hypothesis 3 (several wordings)**.** *On the simple tasks, rule-style lexical task heads that meet Definition 1 under every instruction template, not only under the most accurate one, carry the task more strongly: patched as in E1, they raise accuracy more than the heads of the most accurate template, and the control mean does not raise it more.* Falsified *if, under each readout and in each causal check, their causal effect is larger on at most half of the 14 tasks, or if their larger effects come with a larger effect of the control mean.*

6.1 Design

Question. H3: do the rule-style lexical task heads that every instruction template yields carry the task more strongly than those of the most accurate template? A head that names the task under one template may be copying words of that template; one that names it under all five is less tied to a wording.

Conditions. 14 tasks × the rule style × 2 readouts (Logit-head, J-head). The heads found only by J-head and the example style are not part of E1b.

Procedure.

1. *Identification prompts*: for each of the five instruction templates separately, its first 100 prompts in P^+ (all of them where fewer are answered correctly).
2. *Head sets*: under each readout, a head is in the consensus set if it meets Definition 1 on the prompts of all five templates. The rule-style set of E1 comes from one of the five templates, so the consensus set is contained in it.
3. *Measures*: head count, and the heads the set shares with the E1 set.
4. *Causal checks*: both checks of E1, unchanged (patched items, means of both styles, control mean, α, scoring); only the head sets differ.
5. *Comparison*: per task, readout and check, the causal effect of the consensus set against that of the E1 rule-style set with rule-style means, and in the first check the same for the control mean.

Output. Head sets (Table 9); causal effects of the E1 set and of the consensus set side by side, in the first check (Table 10), with the control mean (Table 11), and in the second check (Table 12). The tables also hold the sets of Sec. 7.

6.2 Results

Head sets (Table 9). The consensus sets ($k = 5$) keep three fifths of the Logit-head heads of E1 and just over half of the J-head heads. They keep least on next-item, singular–plural and park–country under both readouts and on synonym under J-head; synonym has no Logit-head consensus set, so it is not patched.

Causal checks (Tables 10, 11 and 12). The consensus sets give the largest effect of the four sets (E1's and those for $k = 1$, 3 and 5) on tasks where the E1 sets were already effective. In the first check, these are country–capital and park–country under both readouts, product–company under Logit-head and landmark–country under J-head, with 7 to 26 more items than the E1 sets, and product–company under J-head, by less than one item; in the second, country–capital, park–country and product–company under both readouts, with 1 to 5 more items. Elsewhere

the consensus sets are ahead of the E1 sets by less than three items: on prev-item under J-head in the first check, and in the second on English–French and English–Spanish under Logit-head and on country–currency and synonym under J-head. The control mean rises with them by more than three items only at the J-head set of park–country. On most other tasks the consensus sets have a smaller effect than the E1 sets, most in the second check on singular–plural and next-item and under J-head on the translation tasks. On next-item, the J-head set's gain in the first check disappears together with that of the control mean.

Hypothesis. H3 is falsified: under each readout and in each check, the consensus sets raise accuracy more than the E1 sets on at most 5 of the 14 tasks, and by more than three items on at most 3.

	L				J			
Task	E1	$k = 1$	$k = 3$	$k = 5$	E1	$k = 1$	$k = 3$	$k = 5$
antonym	14 (1.4)	30 (2.9)	17 (1.7)	11 (1.1)	21 (2.1)	36 (3.5)	22 (2.1)	13 (1.3)
synonym	2 (0.2)	11 (1.1)	2 (0.2)	0 (0.0)	22 (2.1)	49 (4.8)	21 (2.1)	7 (0.7)
singular–plural	20 (2.0)	38 (3.7)	17 (1.7)	7 (0.7)	24 (2.3)	41 (4.0)	23 (2.2)	8 (0.8)
present–past	10 (1.0)	17 (1.7)	10 (1.0)	4 (0.4)	7 (0.7)	19 (1.9)	7 (0.7)	3 (0.3)
country–capital	21 (2.1)	36 (3.5)	22 (2.1)	14 (1.4)	35 (3.4)	62 (6.1)	36 (3.5)	25 (2.4)
country–currency	17 (1.7)	22 (2.1)	16 (1.6)	14 (1.4)	33 (3.2)	48 (4.7)	31 (3.0)	23 (2.2)
landmark–country	23 (2.2)	38 (3.7)	25 (2.4)	19 (1.9)	27 (2.6)	64 (6.2)	29 (2.8)	15 (1.5)
park–country	24 (2.3)	38 (3.7)	24 (2.3)	11 (1.1)	24 (2.3)	50 (4.9)	25 (2.4)	12 (1.2)
product–company	30 (2.9)	42 (4.1)	28 (2.7)	20 (2.0)	40 (3.9)	74 (7.2)	37 (3.6)	25 (2.4)
next-item	35 (3.4)	45 (4.4)	20 (2.0)	11 (1.1)	47 (4.6)	60 (5.9)	28 (2.7)	13 (1.3)
prev-item	20 (2.0)	36 (3.5)	21 (2.1)	11 (1.1)	35 (3.4)	71 (6.9)	33 (3.2)	20 (2.0)
English–French	11 (1.1)	15 (1.5)	10 (1.0)	10 (1.0)	22 (2.1)	40 (3.9)	18 (1.8)	13 (1.3)
English–German	7 (0.7)	13 (1.3)	9 (0.9)	7 (0.7)	19 (1.9)	34 (3.3)	17 (1.7)	14 (1.4)
English–Spanish	6 (0.6)	8 (0.8)	6 (0.6)	5 (0.5)	14 (1.4)	24 (2.3)	16 (1.6)	8 (0.8)
Mean per task	17.1 (1.7)	27.8 (2.7)	16.2 (1.6)	10.3 (1.0)	26.4 (2.6)	48.0 (4.7)	24.5 (2.4)	14.2 (1.4)
Distinct, all tasks	124 (12.1)	182 (17.8)	112 (10.9)	84 (8.2)	179 (17.5)	270 (26.4)	157 (15.3)	105 (10.3)

Table 9: E1b: rule-style head sets under Logit-head (L) and J-head (J): from the most accurate instruction template (E1, Table 3), and the heads that meet Definition 1 under at least k of the five templates. Each cell: number of heads and, in brackets, its share of the model's 1,024 heads in %. Distinct: heads in the set of at least one task. The set for $k = 1$ contains the E1 set and the set for $k = 5$ is contained in it; of the heads in the sets for $k = 3$, summed over tasks, 198 of 227 (L) and 300 of 343 (J) are in the E1 sets.

	L				J			
Task	E1	$k=1$	$k=3$	$k=5$	E1	$k=1$	$k=3$	$k=5$
antonym	0.04	0.02	0.04	0.02	0.03	0.02	0.02	0.01
synonym	0.01	0.00	0.01	–	0.12	0.21	0.11	0.07
singular–plural	0.03	0.01	0.02	0.00	0.02	0.02	0.01	0.01
present–past	0.00	0.00	0.00	0.00	0.00	0.16	0.00	0.00
country–capital	0.70	0.67	0.68	0.80	0.59	0.60	0.63	0.75
country–currency	0.06	0.06	0.06	0.06	0.12	0.11	0.12	0.12
landmark–country	0.79	0.72	0.72	0.71	0.75	0.75	0.71	0.93
park–country	0.39	0.29	0.34	0.65	0.48	0.42	0.46	0.71
product–company	0.24	0.19	0.23	0.31	0.35	0.35	0.30	0.35
next-item	0.01	0.03	0.00	0.00	0.17	0.22	0.16	0.01
prev-item	0.04	0.07	0.03	0.00	0.04	0.22	0.10	0.06
English–French	0.00	0.00	0.00	0.00	0.02	0.09	0.00	0.00
English–German	0.00	0.00	0.00	0.00	0.01	0.02	0.00	0.00
English–Spanish	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.01

Table 10: E1b, first causal check: causal effect of the rule-style head sets of Table 9 with rule-style means, as in Table 5; 100 patched items per task. –: empty set.

	L				J			
Task	E1	$k=1$	$k=3$	$k=5$	E1	$k=1$	$k=3$	$k=5$
antonym	0.00	0.04	0.04	0.00	0.00	0.00	0.01	0.00
synonym	0.01	0.01	0.01	–	0.02	0.02	0.02	0.05
singular–plural	0.00	0.02	0.00	0.00	0.03	0.02	0.02	0.00
present–past	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
country–capital	0.01	0.01	0.01	0.01	0.00	0.00	0.01	0.01
country–currency	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
landmark–country	0.00	0.00	0.00	0.00	0.09	0.00	0.14	0.02
park–country	0.29	0.24	0.28	0.21	0.32	0.21	0.29	0.43
product–company	0.01	0.01	0.01	0.02	0.00	−0.02	0.01	0.01
next-item	0.00	0.00	0.01	0.00	0.35	0.27	0.34	0.03
prev-item	0.00	0.00	0.00	0.00	0.00	0.02	0.01	0.00
English–French	0.01	0.03	0.01	0.01	0.00	0.06	0.01	0.02
English–German	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
English–Spanish	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 11: E1b, first causal check: causal effect of the control mean at the head sets of Table 9, as the control columns of Table 5.

		L				J			
Task	Items	E1	$k = 1$	$k = 3$	$k = 5$	E1	$k = 1$	$k = 3$	$k = 5$
antonym	100	0.33	0.28	0.31	0.29	0.34	0.36	0.35	0.28
synonym	100	0.04	0.08	0.04	–	0.12	0.10	0.13	0.12
singular–plural	26	0.56	0.58	0.69	0.35	0.72	0.67	0.75	0.42
present–past	61	0.43	0.51	0.40	0.40	0.35	0.64	0.35	0.29
country–capital	27	0.77	0.77	0.81	0.82	0.79	0.81	0.79	0.84
country–currency	65	0.21	0.22	0.19	0.18	0.17	0.17	0.18	0.18
landmark–country	100	0.34	0.31	0.32	0.33	0.32	0.31	0.29	0.31
park–country	45	0.24	0.24	0.24	0.33	0.24	0.24	0.23	0.29
product–company	100	0.14	0.14	0.15	0.16	0.22	0.26	0.21	0.27
next-item	100	0.23	0.32	0.09	0.10	0.45	0.50	0.49	0.17
prev-item	100	0.11	0.12	0.15	0.05	0.14	0.12	0.15	0.09
English–French	100	0.10	0.11	0.10	0.10	0.20	0.25	0.15	0.13
English–German	100	0.09	0.10	0.09	0.09	0.18	0.20	0.15	0.13
English–Spanish	100	0.10	0.13	0.10	0.11	0.27	0.27	0.25	0.19

Table 12: E1b, second causal check: causal effect of the head sets of Table 9 with rule-style means, as in Table 7. Items: the wrongly answered 1-shot prompts patched; the baseline is 0.

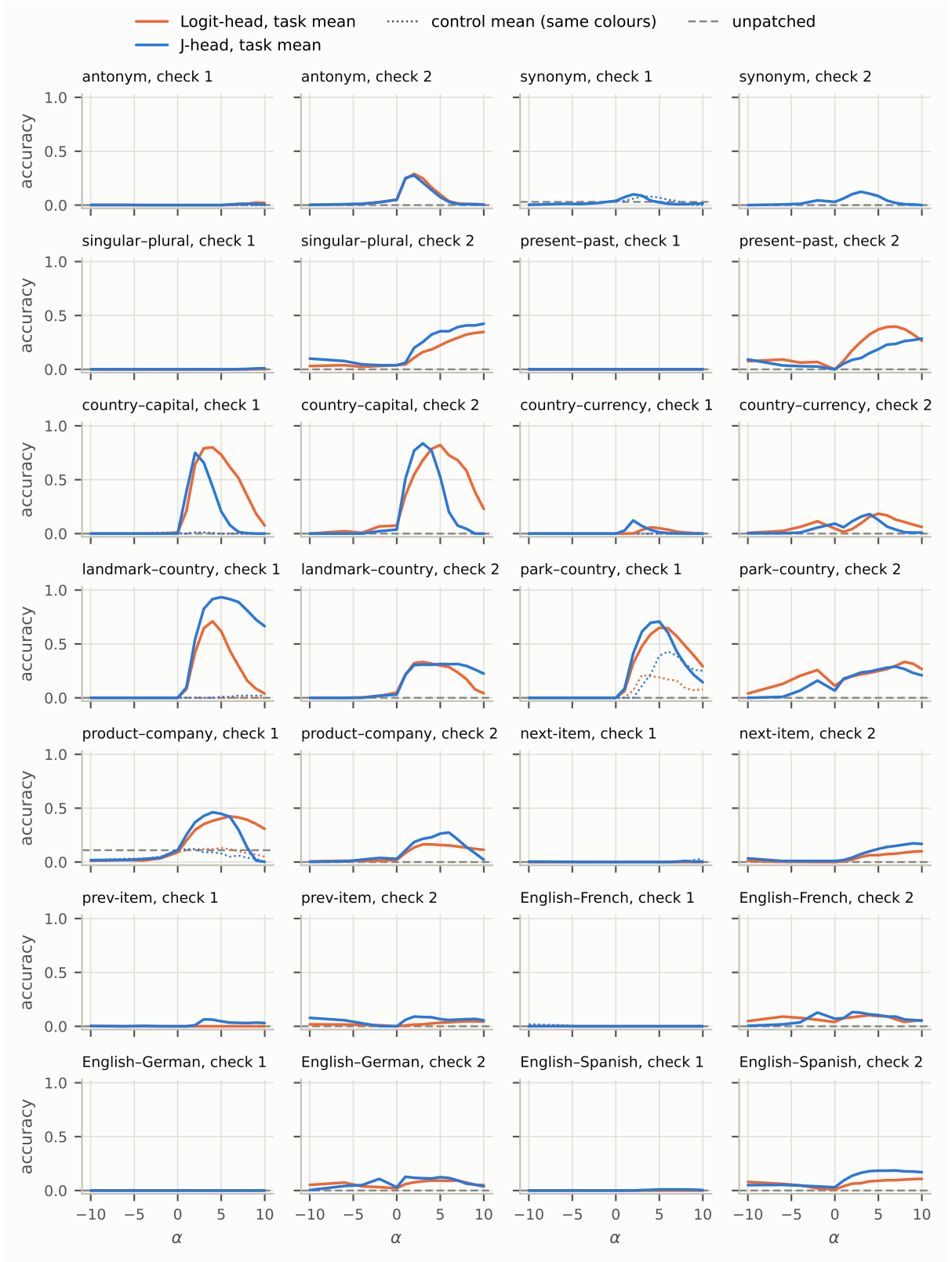


Figure 5: E1b, consensus sets ($k = 5$): accuracy against α for the rule-style sets with rule-style means, averaged over the five instruction templates. For each task, the first check (zero-shot prompts of the first 100 items answered correctly with one shot) and the second (1-shot prompts answered wrongly; no control mean). Solid: the head set's mean over the task's prompts; dotted: the control mean at the same heads; dashed: unpatched accuracy. Synonym has no Logit-head consensus set.

7 E1b, exploratory: heads from at least one or three templates

7.1 Design

Question. After the result of H3, we ask how the causal effect changes when fewer templates are required. This question has no hypothesis, and we report it as exploratory.

Design. As in Sec. 6, with a head in the set for k if it meets Definition 1 on the prompts of at least k of the five templates, for $k \in \{1, 3\}$. The set for $k = 1$ (the union of the five templates' sets) contains the E1 set; the set for $k = 3$ neither contains it nor is contained in it. Head sets and causal effects are in Tables 9–12, beside those of the consensus sets.

7.2 Results

Head sets (Table 9). The three requirements give sets of very different sizes, all small against the model's 1,024 heads: even the union averages under 5% of them per task under J-head, and its sets of all tasks together cover about a quarter. Per task on average, the union ($k = 1$) has about 1.6 times as many heads as the E1 set under Logit-head and 1.8 times under J-head; counted as distinct heads over all tasks, about 1.5 times under both. The sets for $k = 3$ are about as large as the E1 sets, and seven in eight of their heads are E1 heads.

Causal checks (Tables 10, 11 and 12). No requirement gives the largest effect on most tasks; which one does depends on the task.

The union raises J-head's effect on three tasks where the E1 set had little: synonym and prev-item in the first check, and present–past in both, where the first check goes from no effect to 16 items and the second roughly doubles. In the first check, the control mean at these sets stays within two items of zero. The patched present–past prompts are answered with the past tense of the given verb; at the mean and α with the highest accuracy, 94 of the 100 replies differ, so the gain is not one answer repeated. In the first check it is largest at the largest α, 10. Two further gains of the union are not specific to the task: on English–French the control mean rises almost as much, and on next-item the gain stays below that of the control mean. Under Logit-head the union lowers the first check's effect on park–country, landmark–country and product–company, and raises the second check's on next-item and present–past.

The sets for $k = 3$ behave like the E1 sets they mostly consist of. They gain on some tasks and lose on as many, and the union's effect on present–past is gone. Their largest change is on next-item under Logit-head, whose second-check effect falls by more than half; at the J-head set of landmark–country, the control mean rises.

The three requirements thus trade off against each other. Heads that every template yields carry country–capital, park–country and product–company more strongly. Heads that only one or two templates yield add to J-head's effect on present–past, prev-item and synonym, and requiring three templates already removes them. The number of templates under which a head is found is therefore not by itself a measure of how strongly it carries the task.

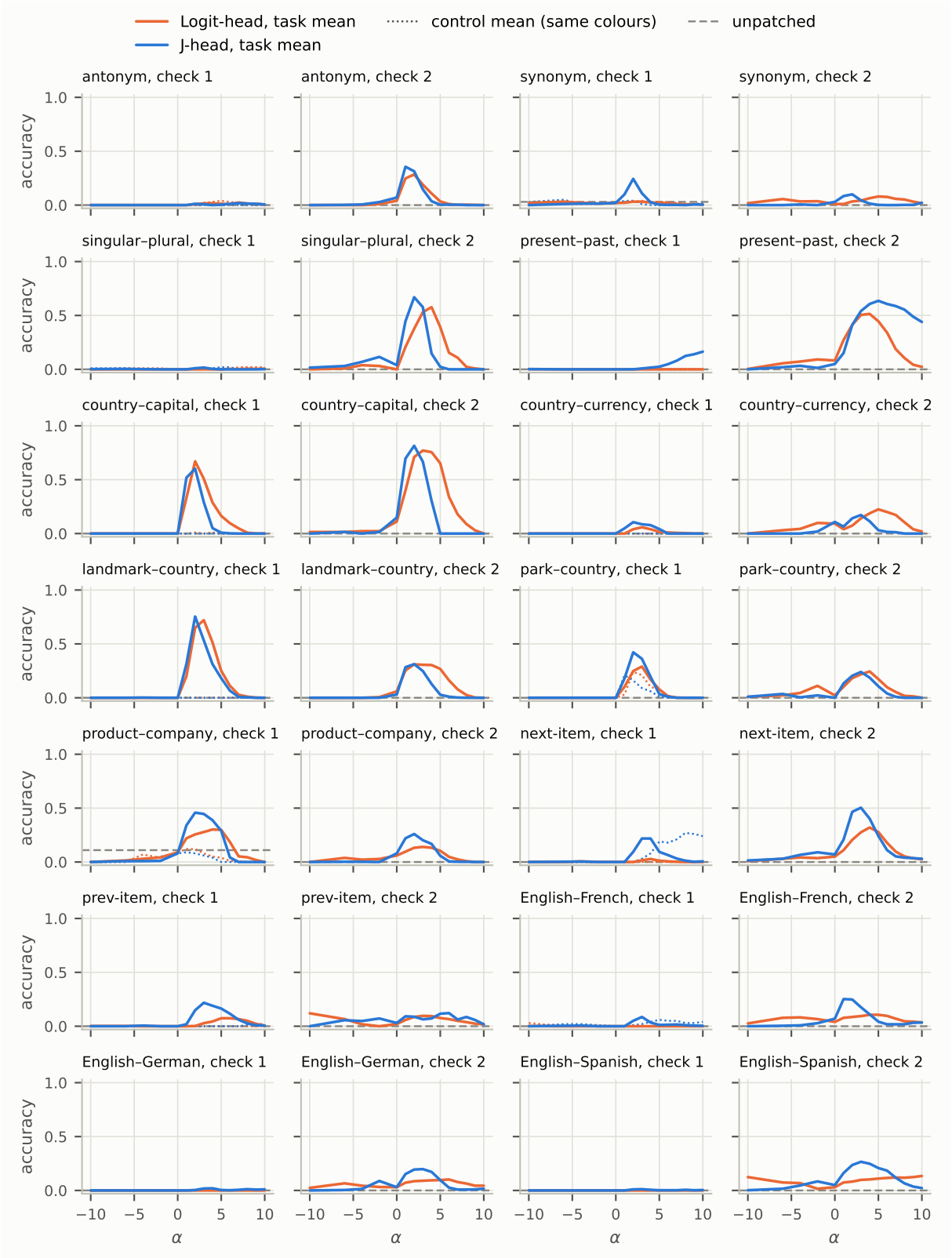


Figure 6: E1b, heads from at least one template ($k = 1$, the union), as Figure 5.

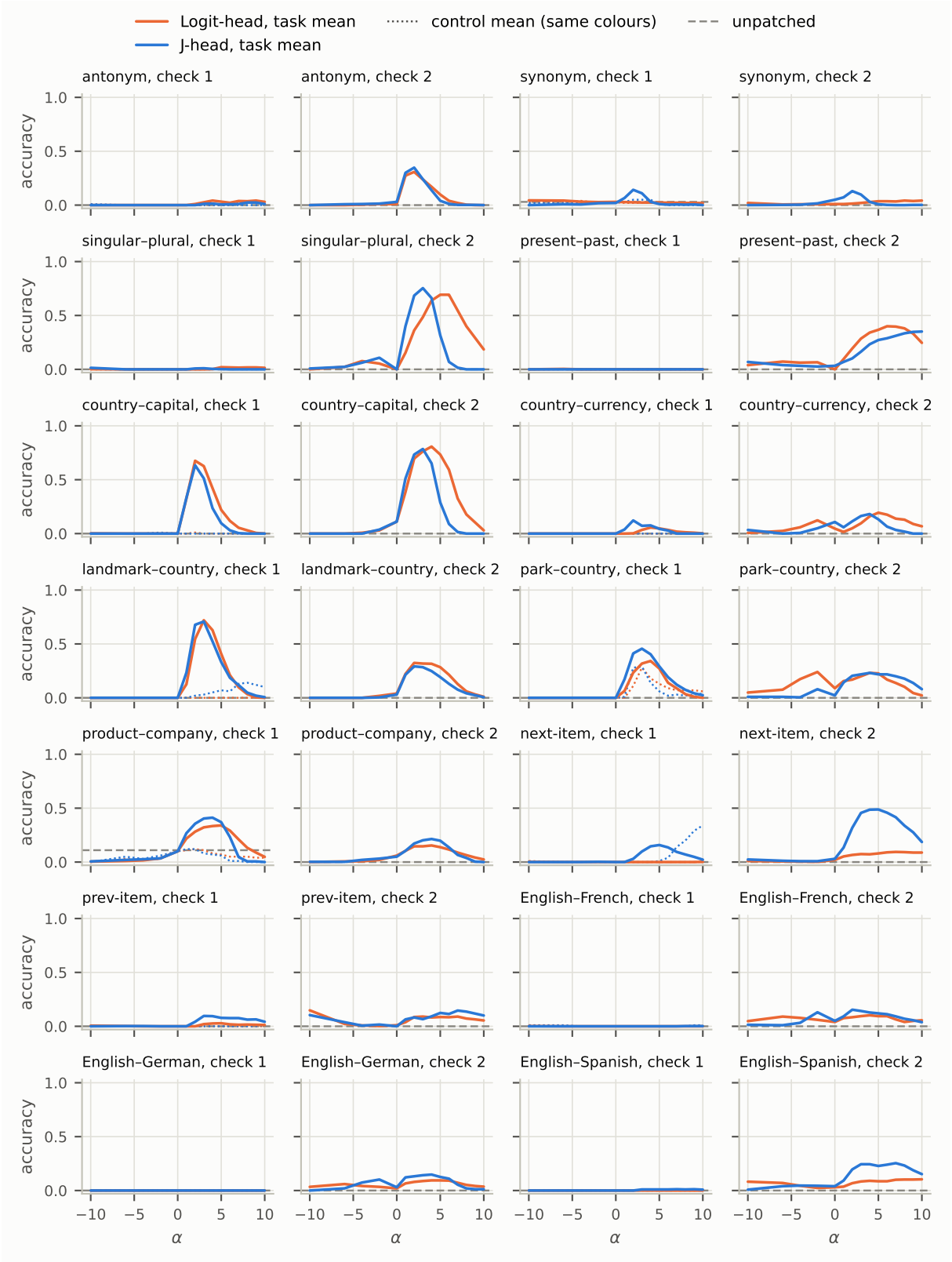


Figure 7: E1b, heads from at least three templates ($k = 3$), as Figure 5.

8 Shared heads of the union

[S1] TO WRITE — Shared heads of the union

J-head $k = 1$ sets restricted to heads in the union sets of at least two tasks; both checks, all tasks (da04856).

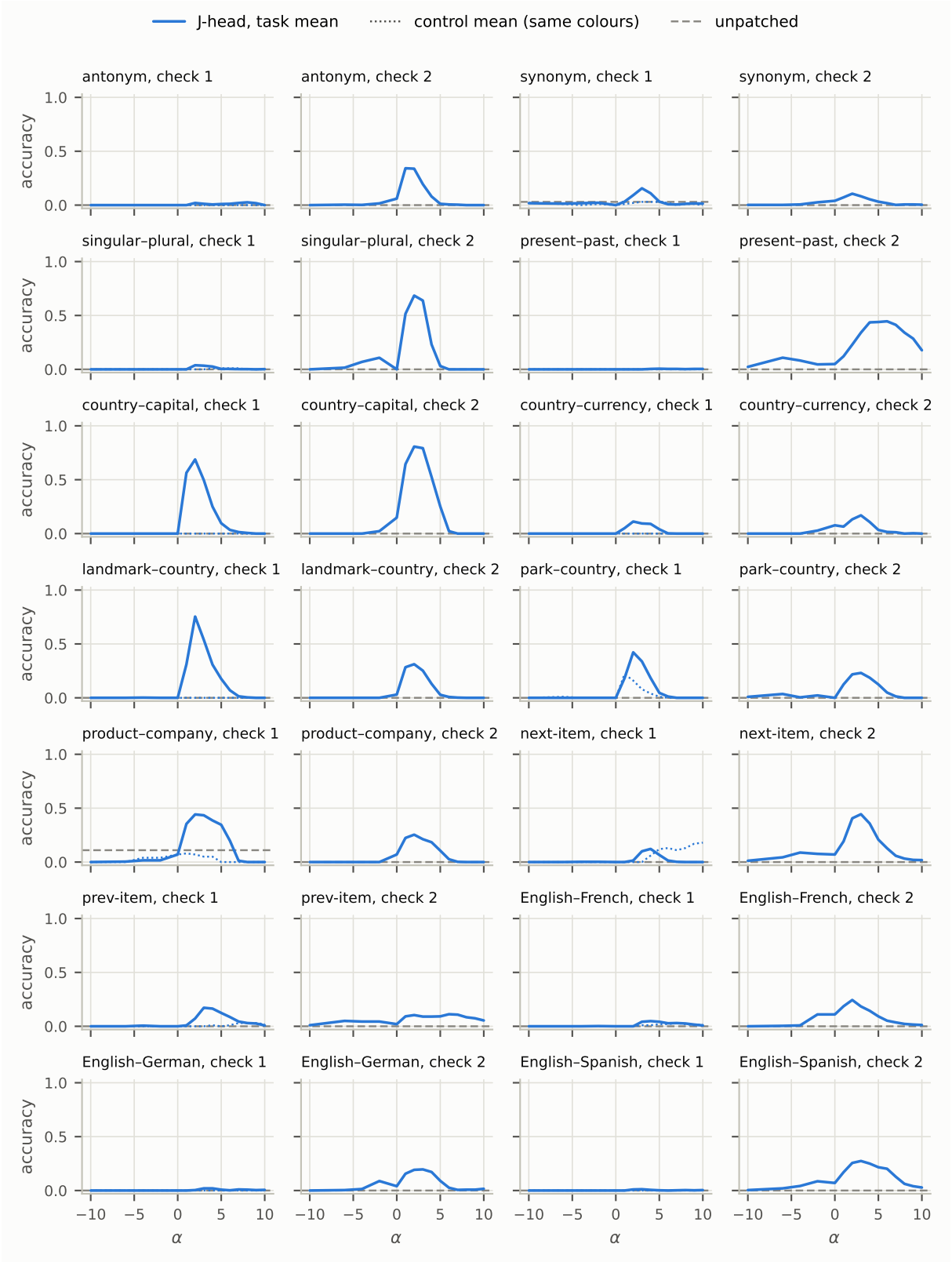


Figure 8: J-head union sets ($k = 1$) restricted to the heads that are also in the union set of at least one other task, as Figure 5.

9 Present–past: the union's four unshared heads

[S2] TO WRITE — Present–past: the union's four unshared heads

The 4 heads only present–past's J-head union set contains (L19H10, L19H24, L20H28, L21H9); both checks, present–past only (c8c6fb8).

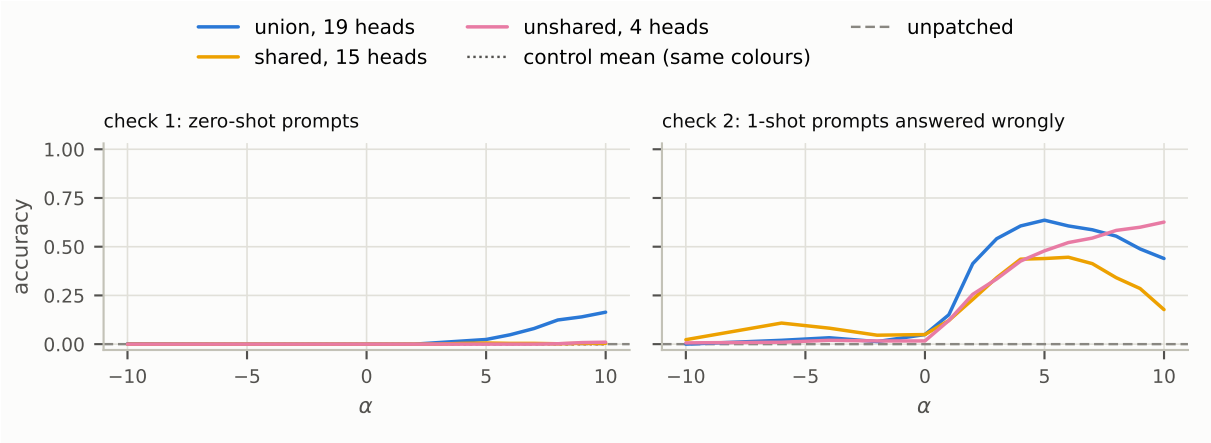


Figure 9: Present–past, J-head, rule-style means averaged over the five instruction templates: the union set ($k = 1$, 19 heads), its 15 heads that are also in another task's union set (shared), and its 4 other heads (unshared: L19H10, L19H24, L20H28, L21H9). Left: first check (100 items); right: second check (61 items, no control mean). Solid: task mean; dotted: control mean at the same heads; dashed: unpatched accuracy.

10 Patched replies by kind

[S3] TO WRITE — Patched replies by kind

Each patched reply sorted by kind, present–past and all tasks; `notebooks/patching_outputs.py` (0239820, 1cfc886).

11 Retrieval heads

[S4] TO WRITE — Retrieval heads

Yang et al.'s retrieval heads (Definition 1 with the answer token as target) and their overlap with every lexical set, against layer-matched chance (1f189a7).

12 Present–past: L21H9 and the other three heads

[S5] TO WRITE — Present–past: L21H9 and the other three heads

L21H9 alone, and L19H10, L19H24, L20H28 alone; both checks (42a8033).

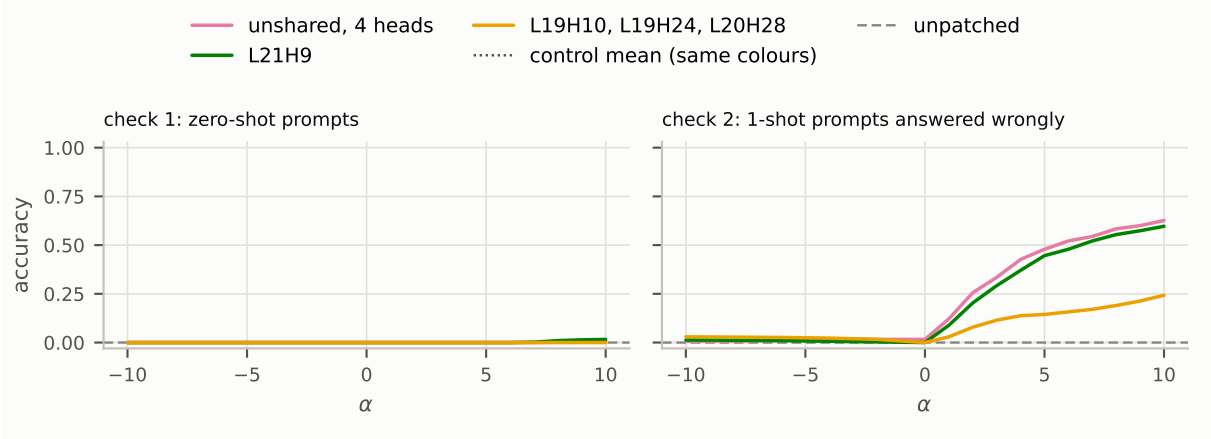


Figure 10: Present–past, as Figure 9: the 4 unshared heads, L21H9 alone, and the other three (L19H10, L19H24, L20H28) alone.

13 The union without retrieval heads

[S6] TO WRITE — The union without retrieval heads

$k = 1$ sets minus every retrieval head of the task; both checks, all tasks (2b0e939).

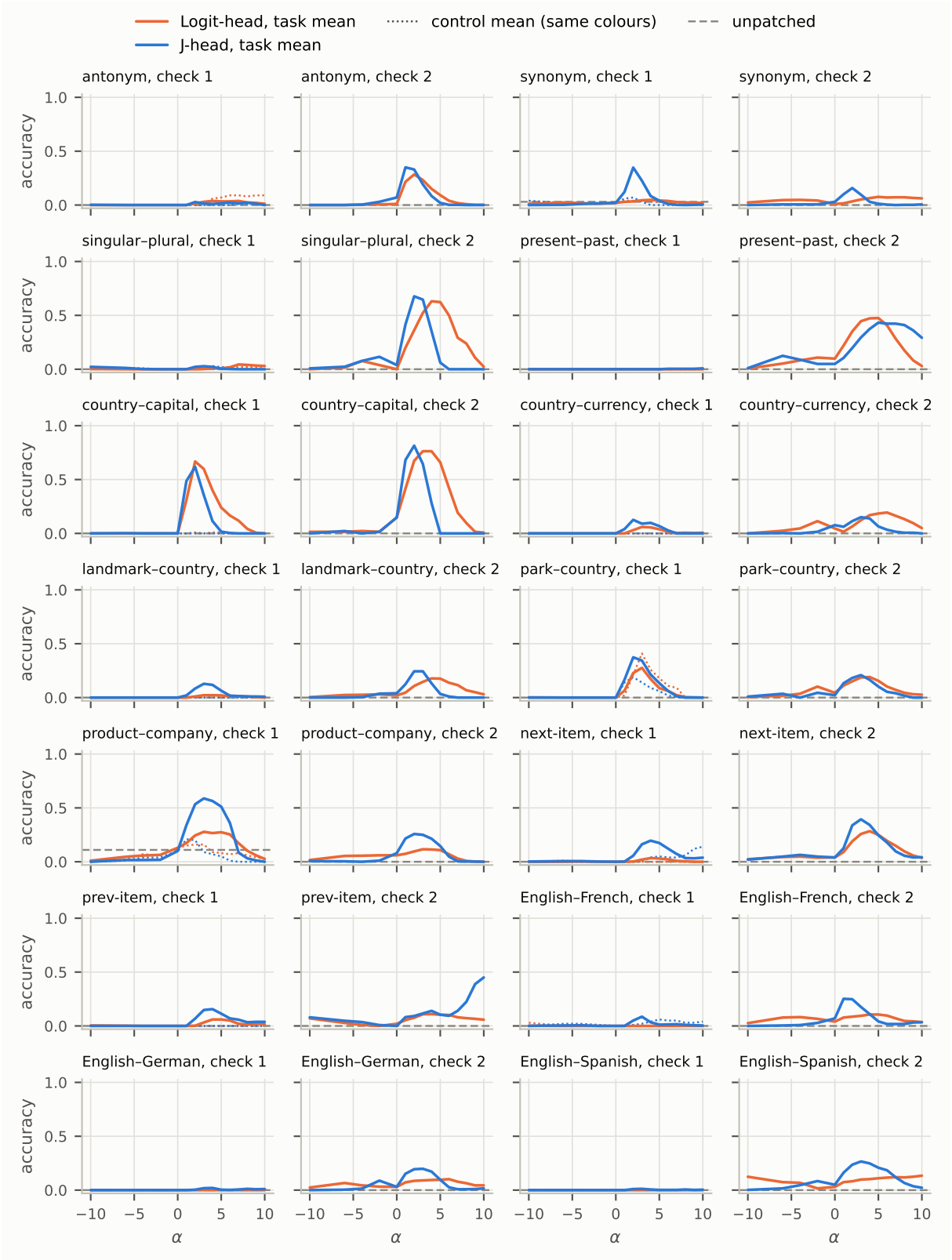


Figure 11: Union sets ($k = 1$) without the task's retrieval heads (found under any template and either readout), as Figure 5.

14 Head sets of equal size

14.1 Design

Question. The head sets compared so far differ in size: the J-head sets are larger than the Logit-head sets on 23 of the 28 pairs of task and style (Sec. 5.2.2), and the union of the five templates' sets has, per task on average, 1.6 to 1.8 times as many heads as the E1 set (Sec. 7.2). A larger set changes more of the forward pass, so part of a difference in causal effect may come from the number of heads rather than from which heads are chosen. Do the differences between readouts and between template rules remain when every set has the same number of heads? The question was asked after the results of the earlier sections; it has no hypothesis, and we report it as exploratory.

Conditions. 14 tasks $\times$ 2 readouts $\times$ 3 sizes, $N \in \{5, 10, 20\}$, $\times$ 4 selection rules: the rule of E1 (the most accurate instruction template for the rule style, 5 shots for the example style; both styles), and for the rule style the rules of E1b (at least k of the five templates, $k \in \{1, 3, 5\}$).

Procedure.

1. *Ranking*: under each rule and readout, every head of the model is ranked by the quantity that Definition 1 thresholds: its match rate on E1's identification prompts (E1), or the k-th highest of its match rates under the five templates (E1b). Ties are broken by the mean number of term tokens among the head's ten top tokens on the same prompts, then by layer and head. Recomputed from these rates, the threshold sets are those of E1 and E1b.

2. *Head sets*: the first N heads. Only heads that match on at least one prompt are ranked; where fewer than N do, the set of size N is not formed. This leaves out 37 of the 420 sets, all of size 10 or 20 except the J-head consensus set of present–past for $N = 5$, most of them Logit-head sets of the translation tasks or consensus sets.

3. *Measures*: how many of the N heads meet the rule's threshold, and how many the Logit-head and J-head sets of the same rule and size share.

4. *Causal checks*: both checks of E1 (patched items, control mean, scoring), each set patched with the means of its own style only, at nine of the fifteen values of α: -10, -4, 2, 3, 4, 5, 6, 8 and 10. On the threshold sets of E1 and E1b, the highest accuracy over these nine values is within 0.05 of that over all fifteen, and within 0.02 on all but 4 of the 417 combinations of set, check and mean; for the comparison, the effects of the threshold sets are taken over the same nine values. A set equal to another of the same style and task is patched once.

5. *Comparison*: the causal effect at equal size, in each check: J-head against Logit-head for each task, style, rule and N; each E1b rule against E1 for each task, readout and N. A difference seen between the threshold sets holds at equal size on a task if the same rule or readout has the larger effect at every N for which both sets are formed.

Output. Head sets and measures (`results/sized`); causal effects at each size; accuracy against α.

14.2 Results

Head sets. Where a threshold set has at least N heads, the set of size N is its N highest-ranked heads; where it has fewer, the set adds the highest-ranked heads below the threshold.

Below-threshold heads are therefore rare at size 5 and more frequent at size 20, more so under Logit-head, whose threshold sets are smaller. At each size, the Logit-head and J-head sets of the same rule share about three in five of their heads.

Readouts (Table 13, Figures 12–18; effects in Tables 14–16, App. A). J-head's advantage over Logit-head does not come from its larger sets. Taking the differences between the threshold sets of at least 0.05 (a descriptive cut, not part of the pre-stated comparison), J-head had the larger effect on most of them, in both styles and both checks, and on about two thirds of these it is larger at every size as well; on none is Logit-head. The few tasks on which Logit-head's threshold set had the larger effect do not keep it: on country–capital (rule style, first check) and present–past (rule style, second check) J-head is larger at every size. On country–capital, J-head's threshold set has more than one and a half times as many heads as Logit-head's, and its effect falls with the number of heads; on landmark–country and park–country (example style, first check) the order changes with the size.

Template rules (Table 13, Figures 19–26; effects in Tables 17 and 18, App. A). Most differences between E1 and the rules of E1b change direction with the size, so they come from how many heads a rule selects, or from single heads, rather than from the rule. The few that hold at equal size are mostly about the consensus heads. In the second check, the consensus heads ($k = 5$) have smaller effects than the E1 heads of the same number on several tasks, under J-head (antonym, singular–plural, next-item, English–German) and on singular–plural under Logit-head; few consensus sets of size 20 are formed, so this rests mostly on sizes 5 and 10. In the first check, the consensus heads have the larger effect on park–country under both readouts, a task on which the control mean also has large effects (below). The union's gain on present–past under J-head (Sec. 7.2) holds as well, but at equal size it appears only at size 20 (below).

Size. The effect does not grow steadily with the number of heads. On country–capital in the rule style, the five highest-ranked heads give the largest effect under both readouts; on next-item and prev-item in the second check, and on singular–plural in the example style, the sets of size 20 give the largest. Some single sizes stand out against both neighbours, such as the Logit-head example set of antonym at size 20 in the first check, which is the only antonym set with a large effect.

One head on present–past. Under J-head with example-style heads, the first check has no effect at the threshold set, which has four heads, and a large one at sizes 5 and 10, with no effect of the control mean. The fifth head added, L15H1, matches on 7% of the identification prompts, below the threshold. It is also in the J-head union set of size 20 for the rule style, the only rule-style set with an effect on present–past in the first check, and in no other set of the task. The patched replies begin with the past tense of the given verb and differ from item to item. That L15H1 carries the effect is so far a comparison across sets; patching it alone, and the sets without it, would test it.

Control mean (dotted curves in Figures 13, 15 and 17). On park–country the control mean has large effects at most sizes, in some sets as large as those of the task mean. The patched model answers with a title that begins with the country's name, such as "Albania's Hidden Gem", which the flexible score accepts. The control mean also comes close to or exceeds the task mean at the J-head example sets of singular–plural (size 5) and next-item (size 20). Effects on these tasks are not specific to the task.

Thus, at equal size, the difference between the readouts largely remains, while the differences between template rules mostly do not.

	First check		Second check	
Pair (first vs second)	first larger	second larger	first larger	second larger
J-head vs Logit-head, rule style	9 / 3 / 0	5 / 1 / 1	11 / 7 / 0	3 / 0 / 1
J-head vs Logit-head, example style	10 / 5 / 1	4 / 0 / 0	9 / 6 / 0	4 / 0 / 1
$k = 1$ vs E1, Logit-head	3 / 0 / 0	8 / 1 / 0	10 / 2 / 1	3 / 1 / 0
$k = 3$ vs E1, Logit-head	1 / 0 / 0	8 / 1 / 0	6 / 2 / 0	6 / 1 / 0
$k = 5$ vs E1, Logit-head	3 / 1 / 1	6 / 3 / 0	5 / 0 / 0	7 / 3 / 1
$k = 1$ vs E1, J-head	9 / 1 / 0	5 / 1 / 0	7 / 3 / 0	6 / 0 / 0
$k = 3$ vs E1, J-head	3 / 0 / 0	11 / 2 / 1	5 / 0 / 0	7 / 1 / 0
$k = 5$ vs E1, J-head	5 / 2 / 0	9 / 3 / 0	5 / 0 / 1	9 / 4 / 0

Table 13: Equal-size sets: for each pair and check, the number of tasks on which the first or the second had the larger effect between the threshold sets, and of these, on how many the same one is larger at every size for which both sets are formed (holds) and on how many the other one is (reverses). Cells: tasks / holds / reverses. Effects of the threshold sets over the nine values of α of the procedure; ties and tasks without both threshold sets are counted on neither side. Every difference is counted, however small.

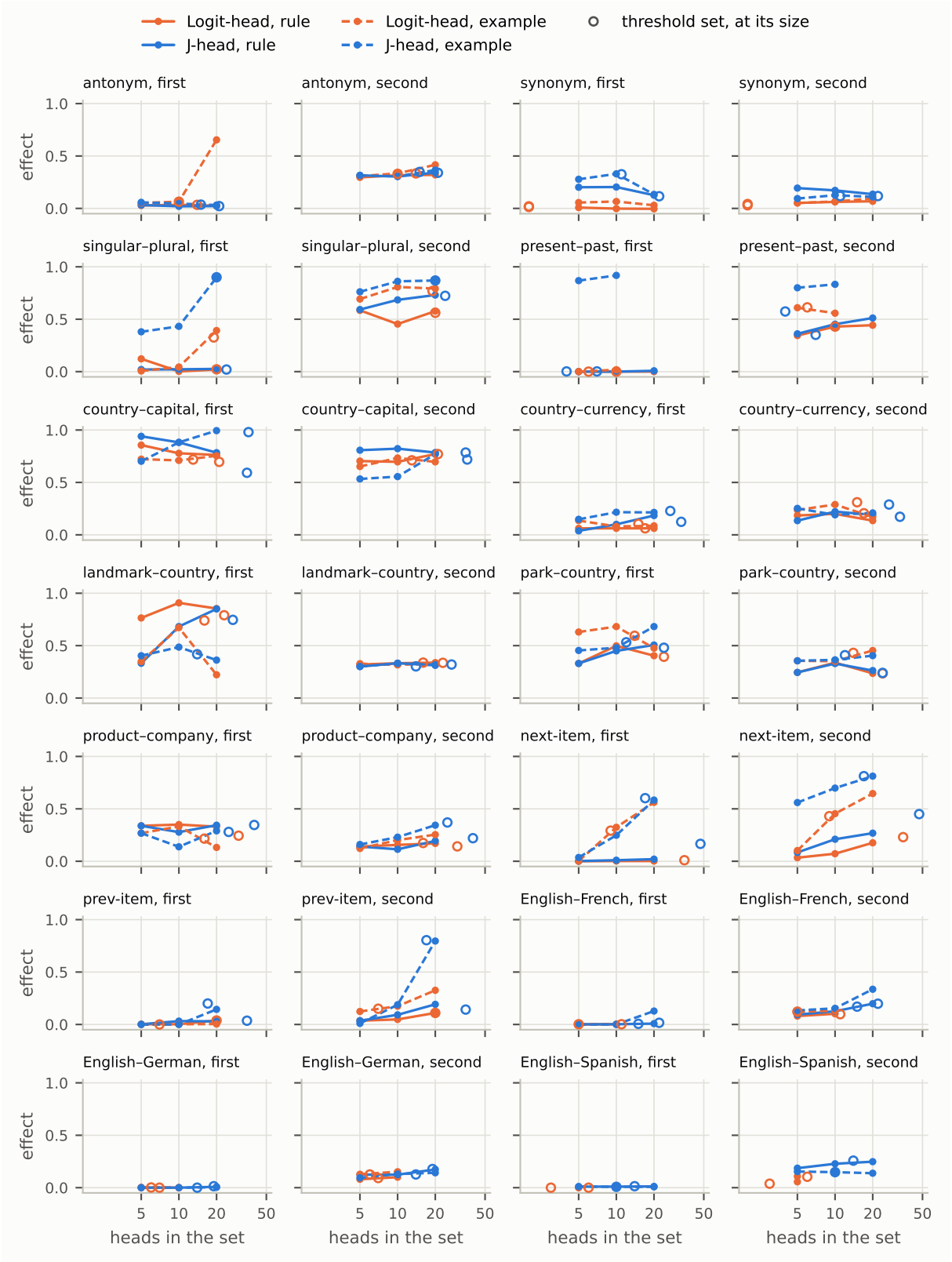


Figure 12: Equal-size sets, E1 rule: causal effect against the number of heads (log scale), per task, in the first and the second check. Filled markers: sets of size 5, 10 and 20; hollow markers: the threshold set, at its own size. Solid: rule style; dashed: example style.

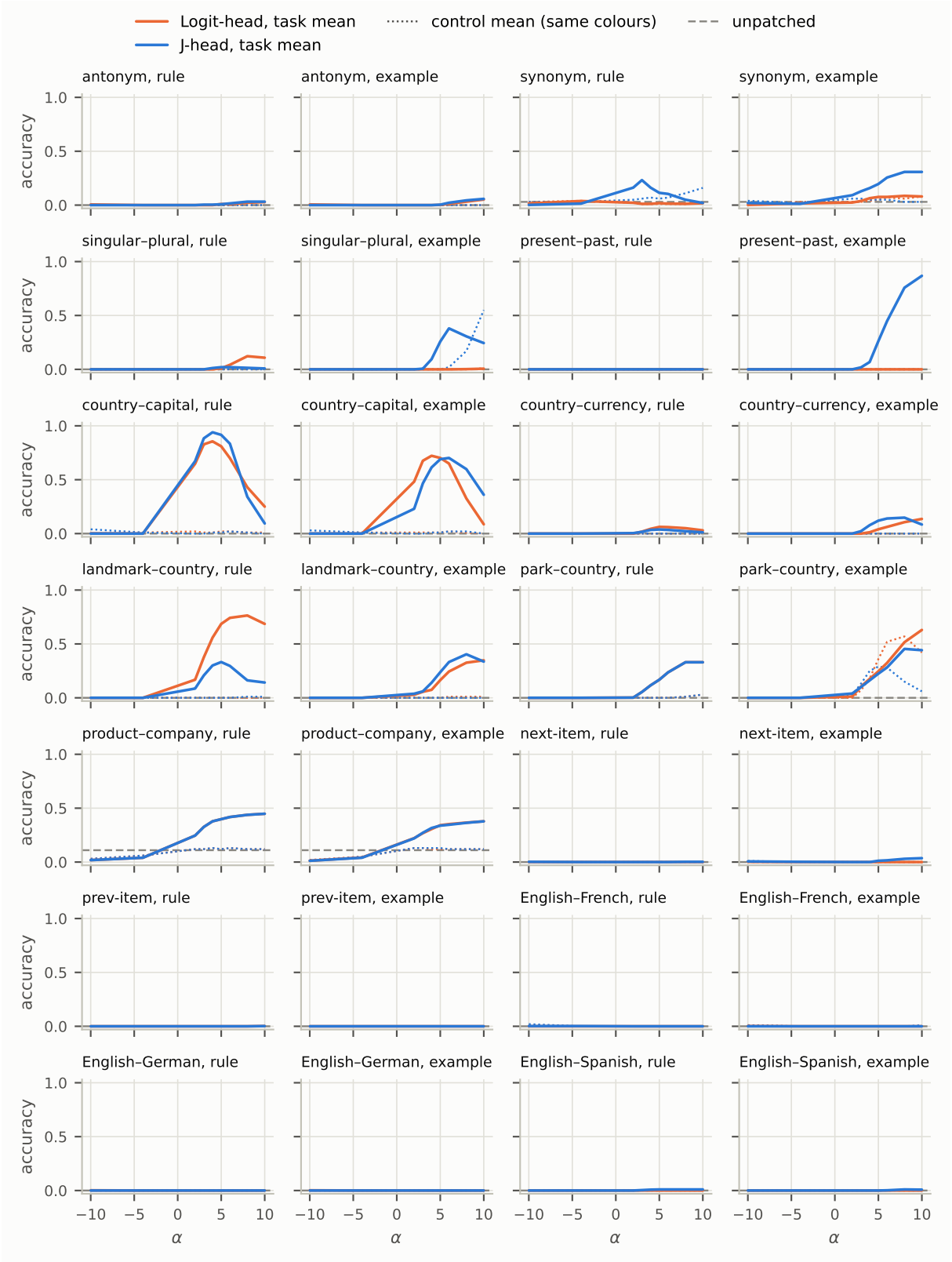


Figure 13: Equal-size sets of the 5 highest-ranked heads under the rule of E1, first causal check: accuracy on the patched items against α, with each style's set patched with means of the same style, averaged over the style's five source wordings, as Figure 3. Nine values of α: -10, -4, and 2 to 10 (none between -4 and 2). Solid: the task mean; dotted: the control mean at the same heads; dashed: unpatched accuracy. Empty panel: fewer than 5 heads match on any prompt.

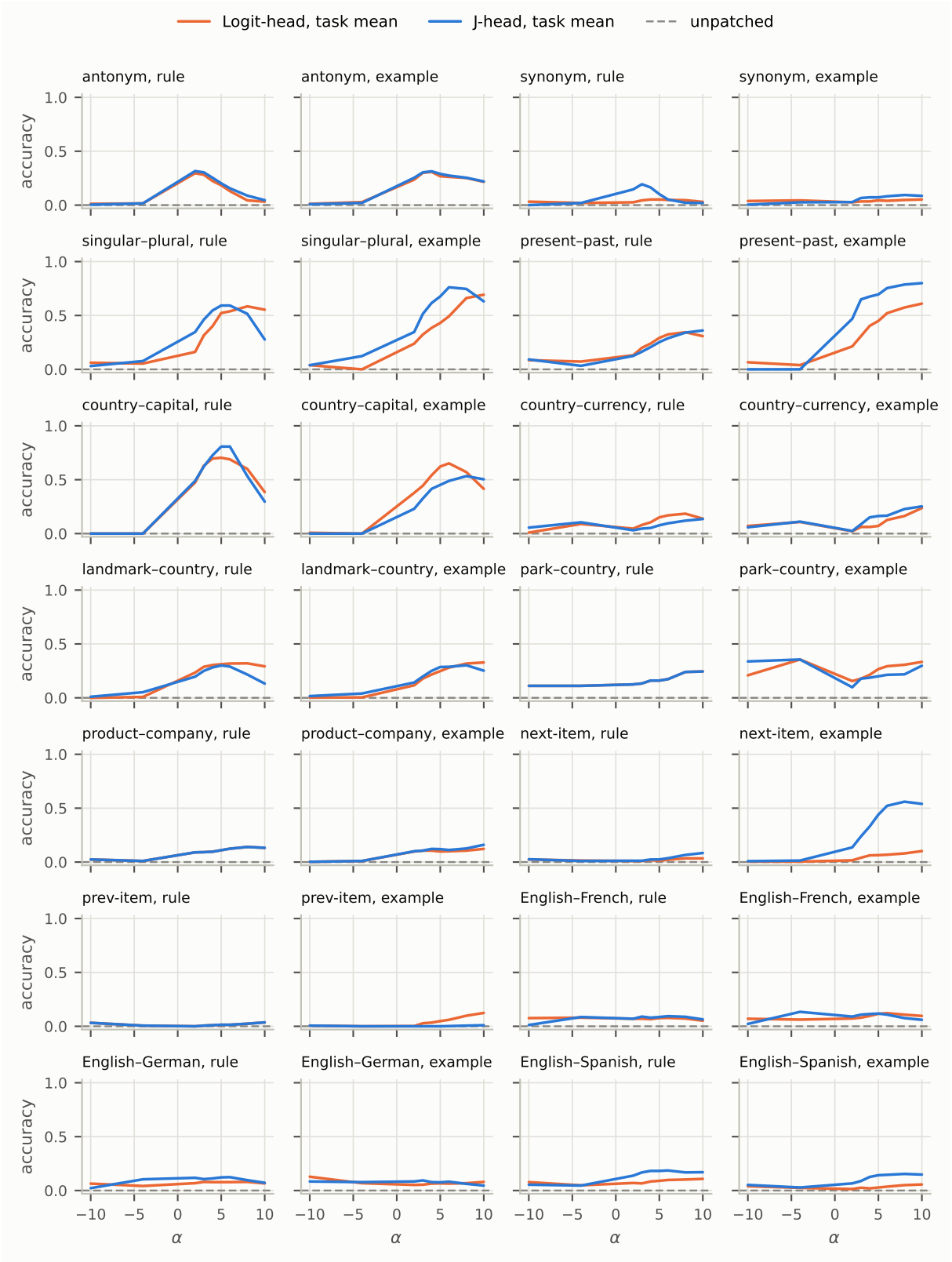


Figure 14: Equal-size sets of 5 heads, E1 rule, second causal check, as Figure 13, with the wrongly answered 1-shot prompts patched. No control mean.

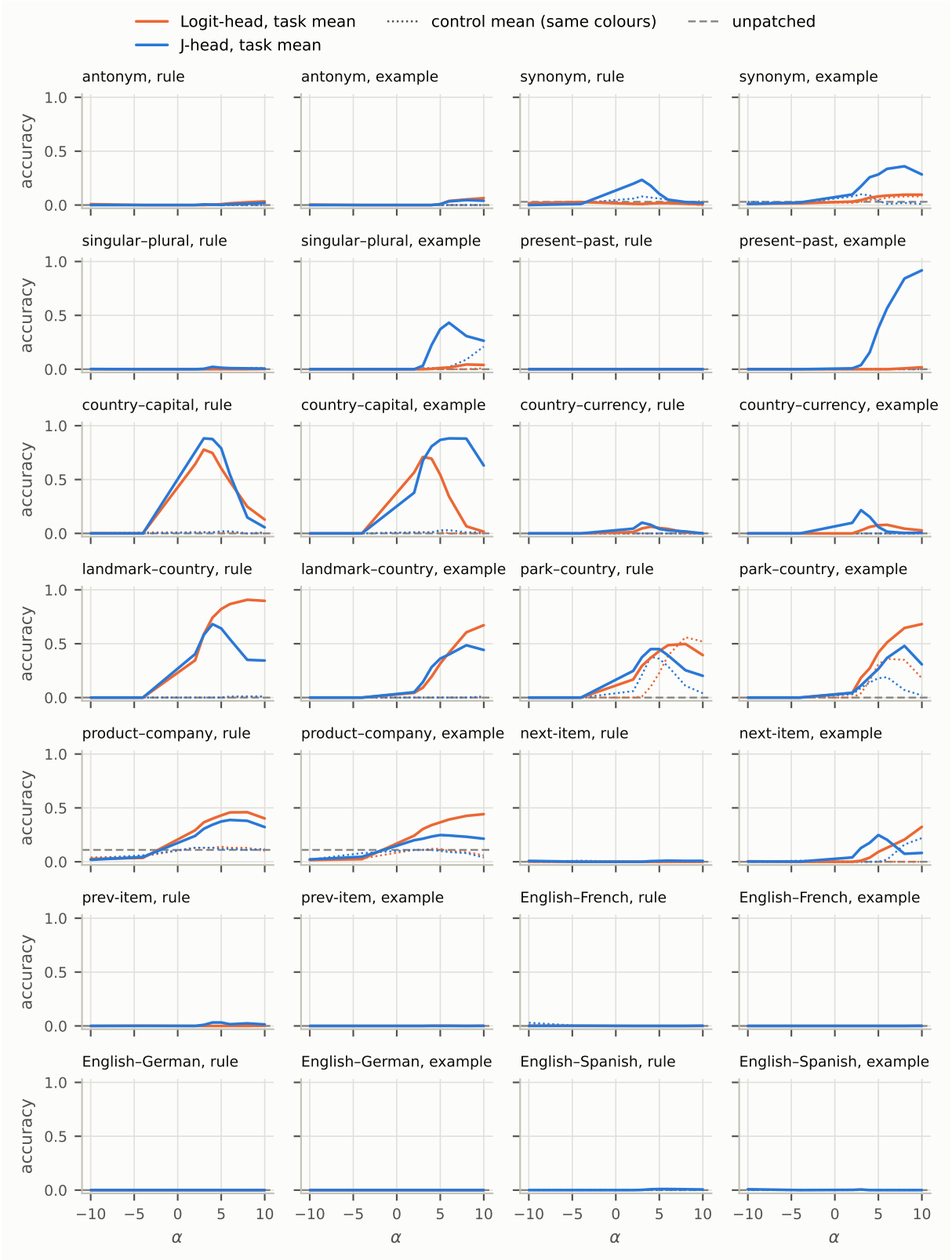


Figure 15: Equal-size sets of 10 heads, E1 rule, first causal check, as Figure 13.

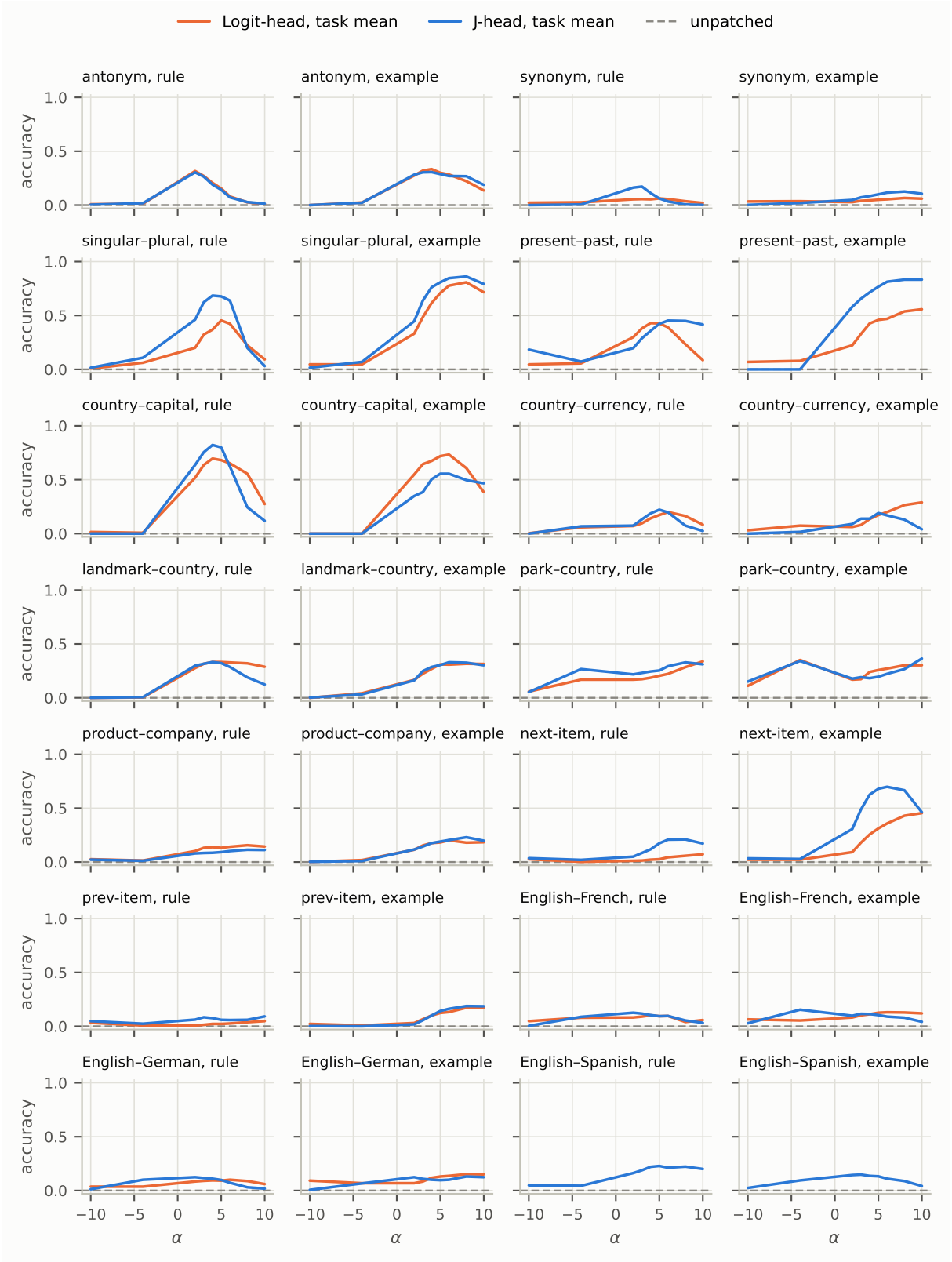


Figure 16: Equal-size sets of 10 heads, E1 rule, second causal check, as Figure 14.

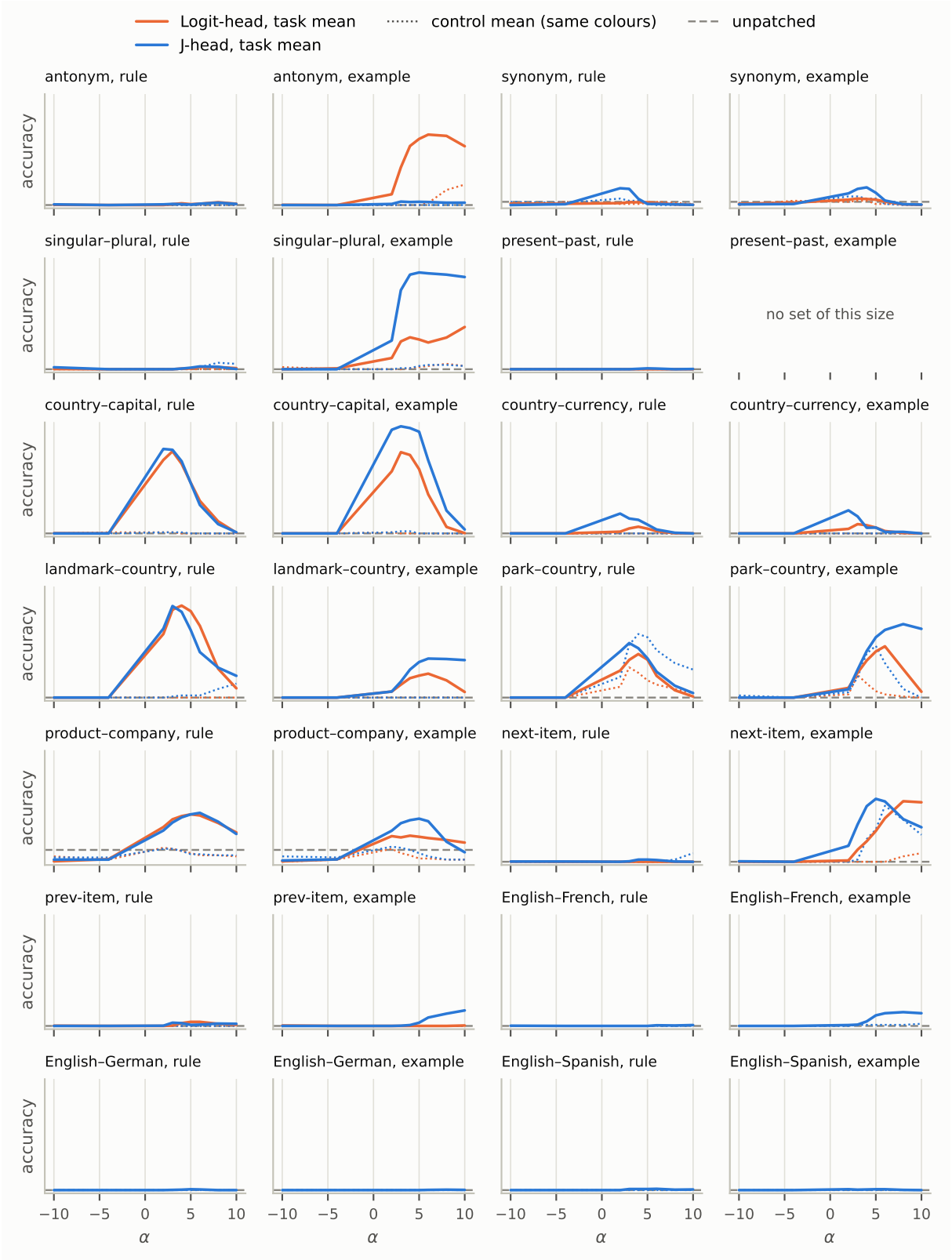


Figure 17: Equal-size sets of 20 heads, E1 rule, first causal check, as Figure 13.

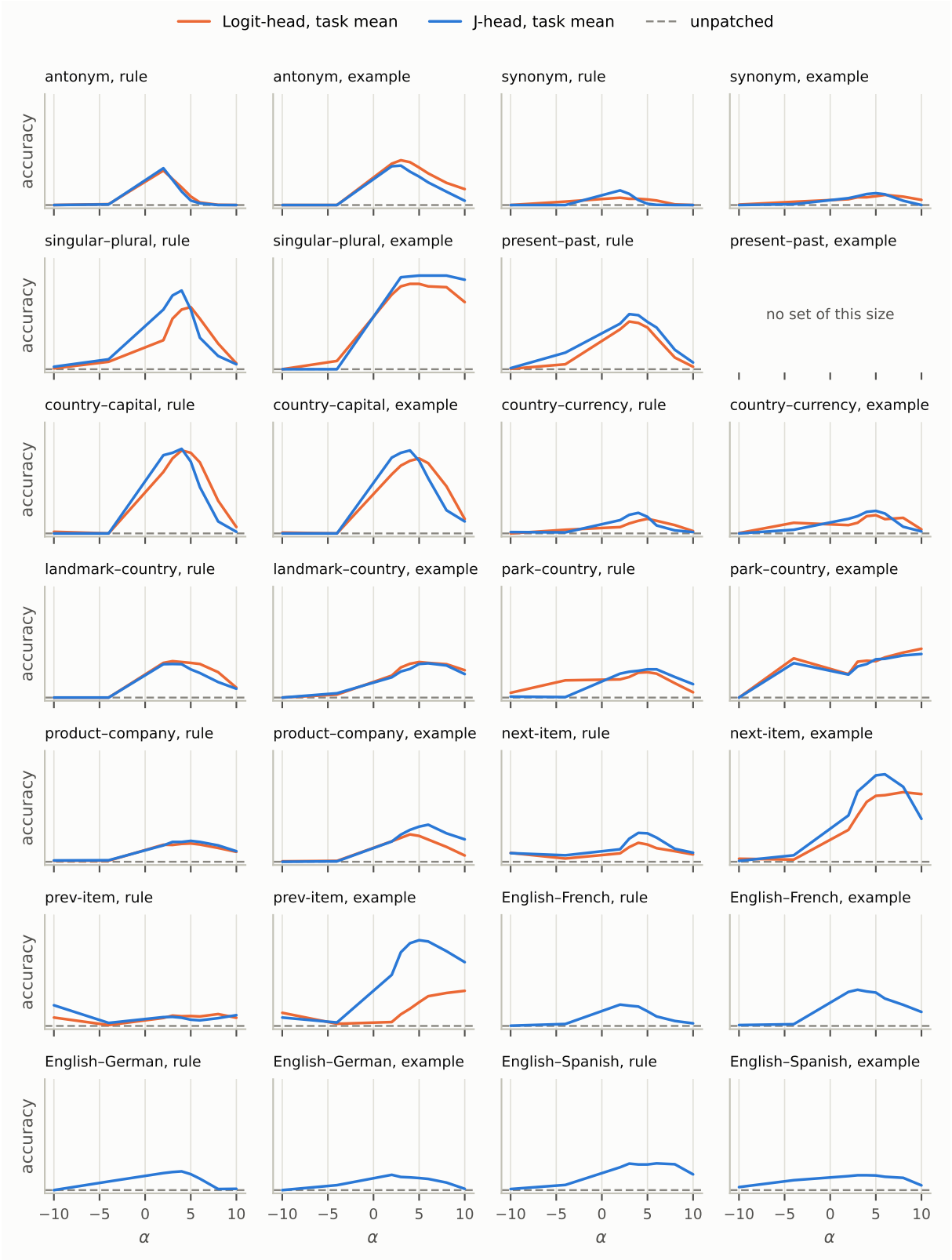


Figure 18: Equal-size sets of 20 heads, E1 rule, second causal check, as Figure 14.

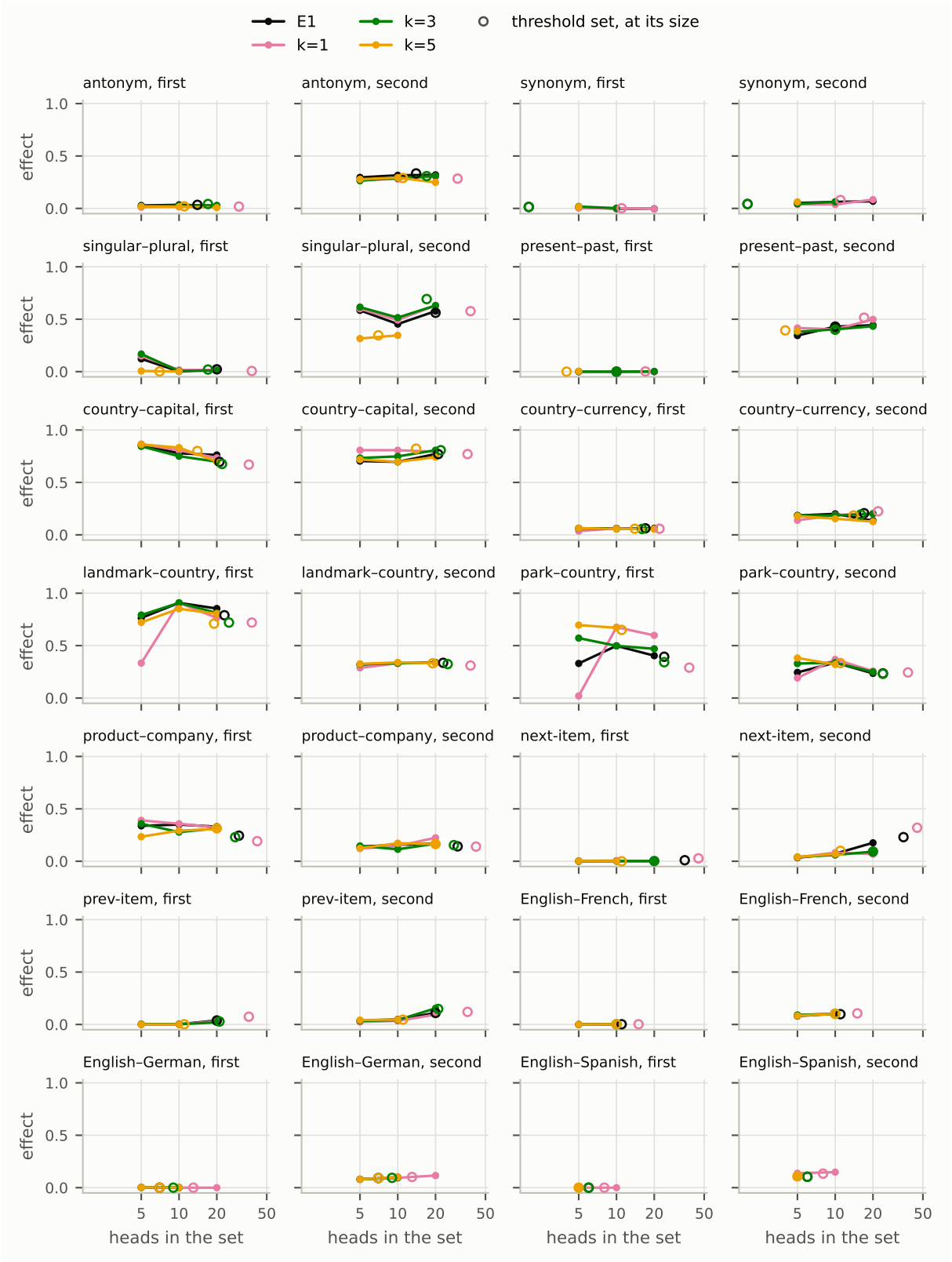


Figure 19: Equal-size sets, rule style, Logit-head: causal effect against the number of heads under the rule of E1 and the rules of E1b, as Figure 12.

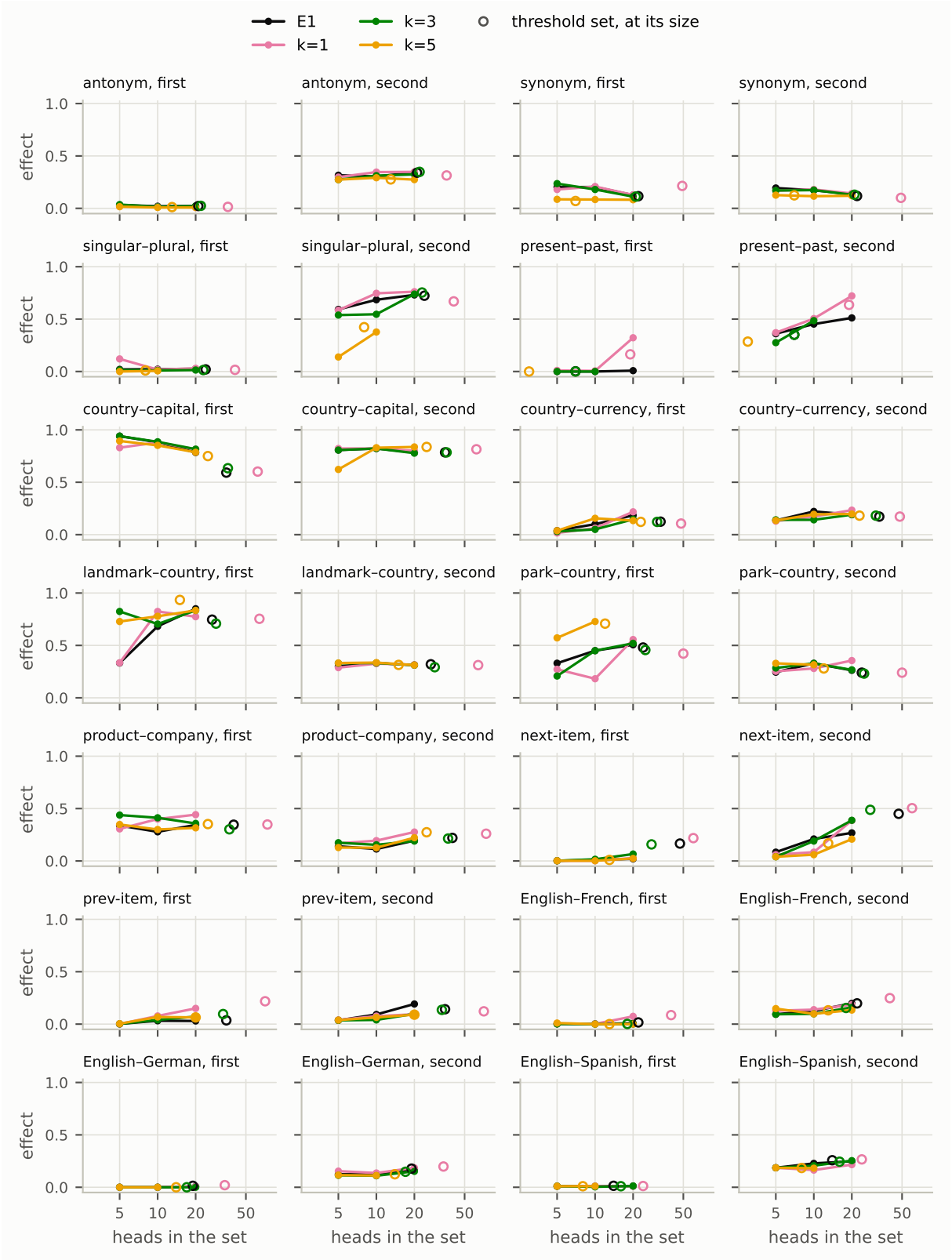


Figure 20: Equal-size sets, rule style, J-head, as Figure 19.

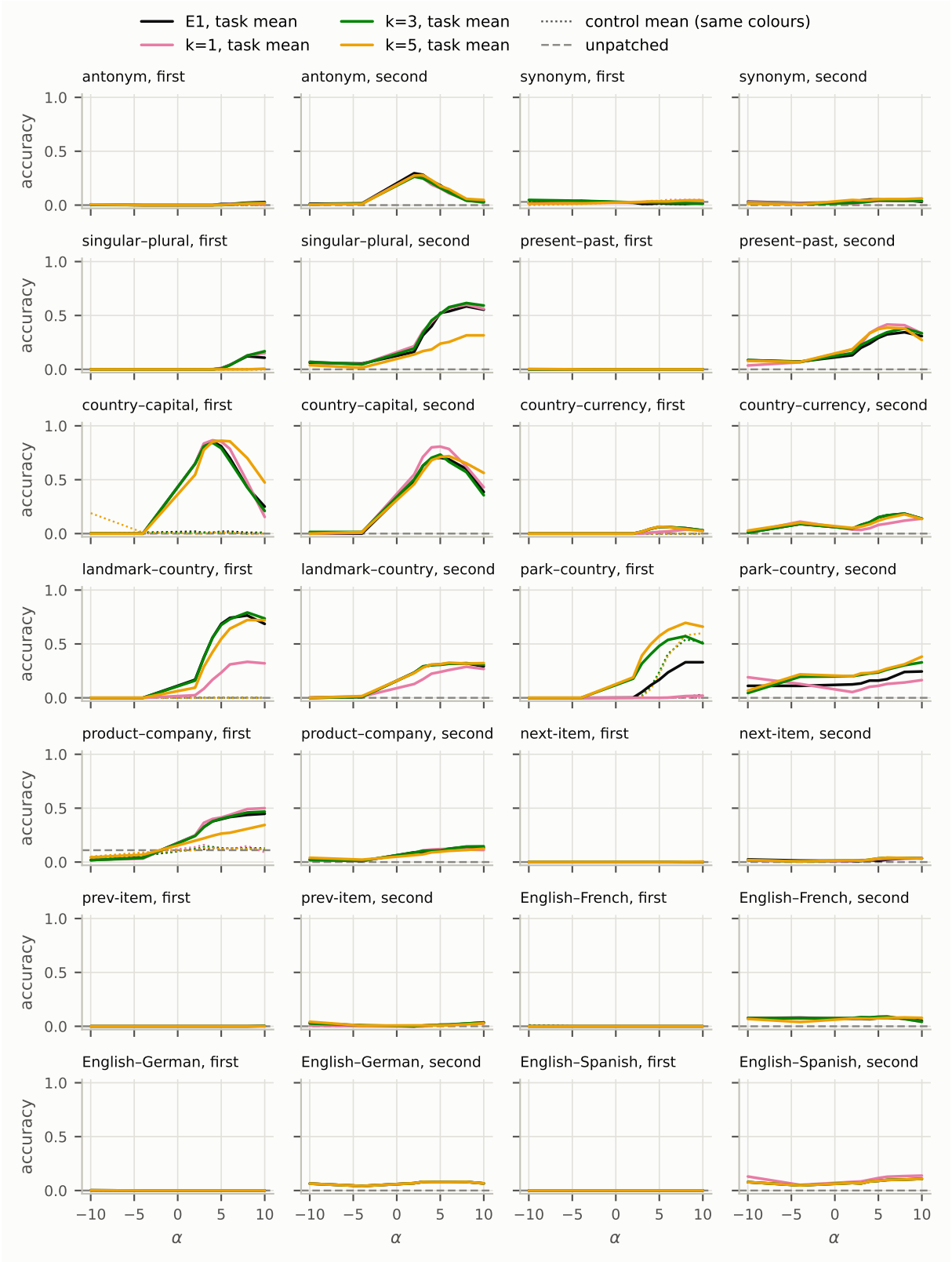


Figure 21: Equal-size sets of the 5 highest-ranked Logit-head heads, rule style, under the rule of E1 and the rules of E1b (at least k of the five instruction templates): accuracy against α with rule-style means, averaged over the five templates, in the first check (with the control mean, dotted) and the second, as Figure 5, at the nine values of α of Figure 13. A missing curve: fewer than 5 heads match under that rule.

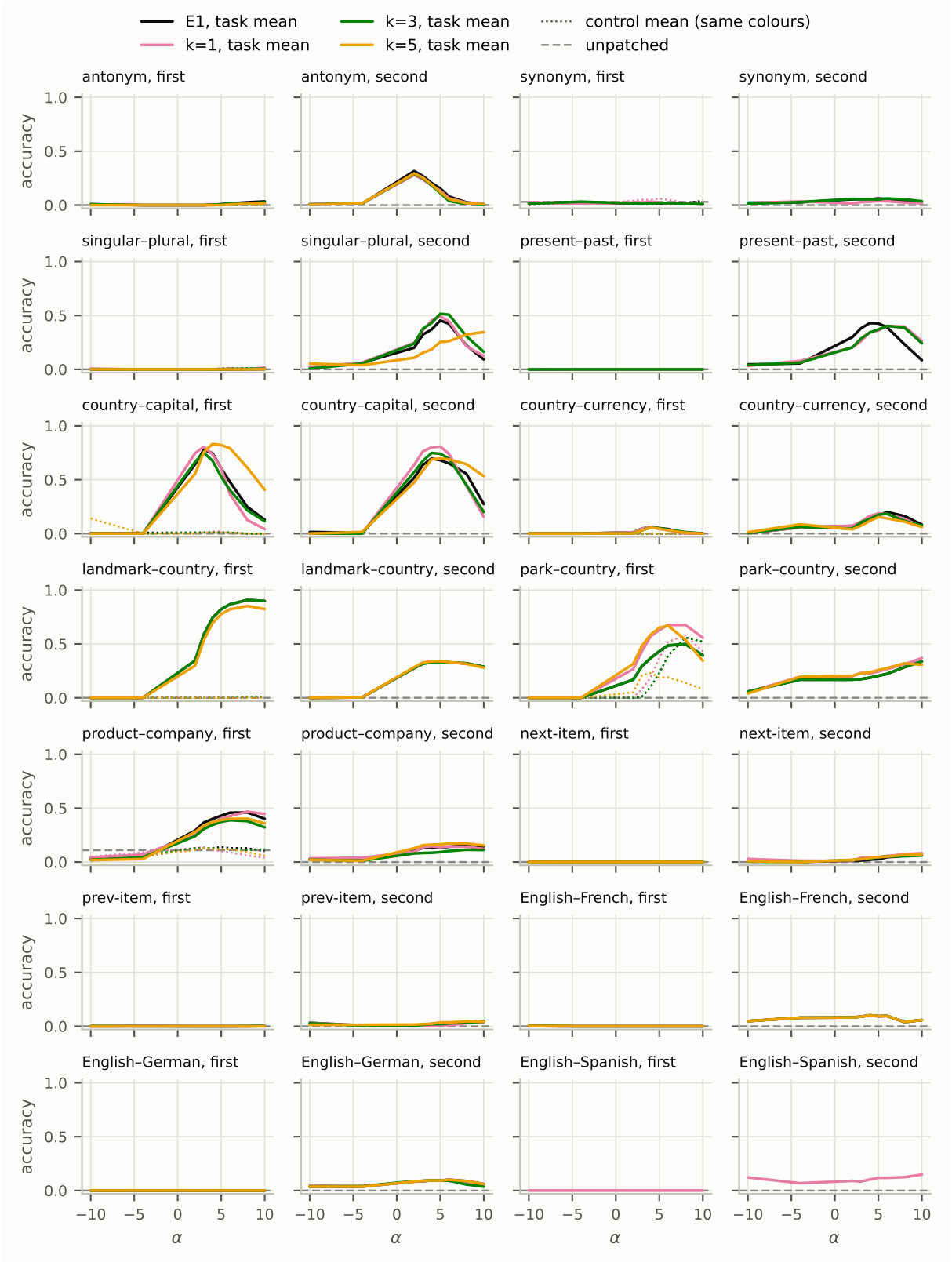


Figure 22: Equal-size sets of 10 Logit-head heads, rule style, E1 and E1b rules, as Figure 21.

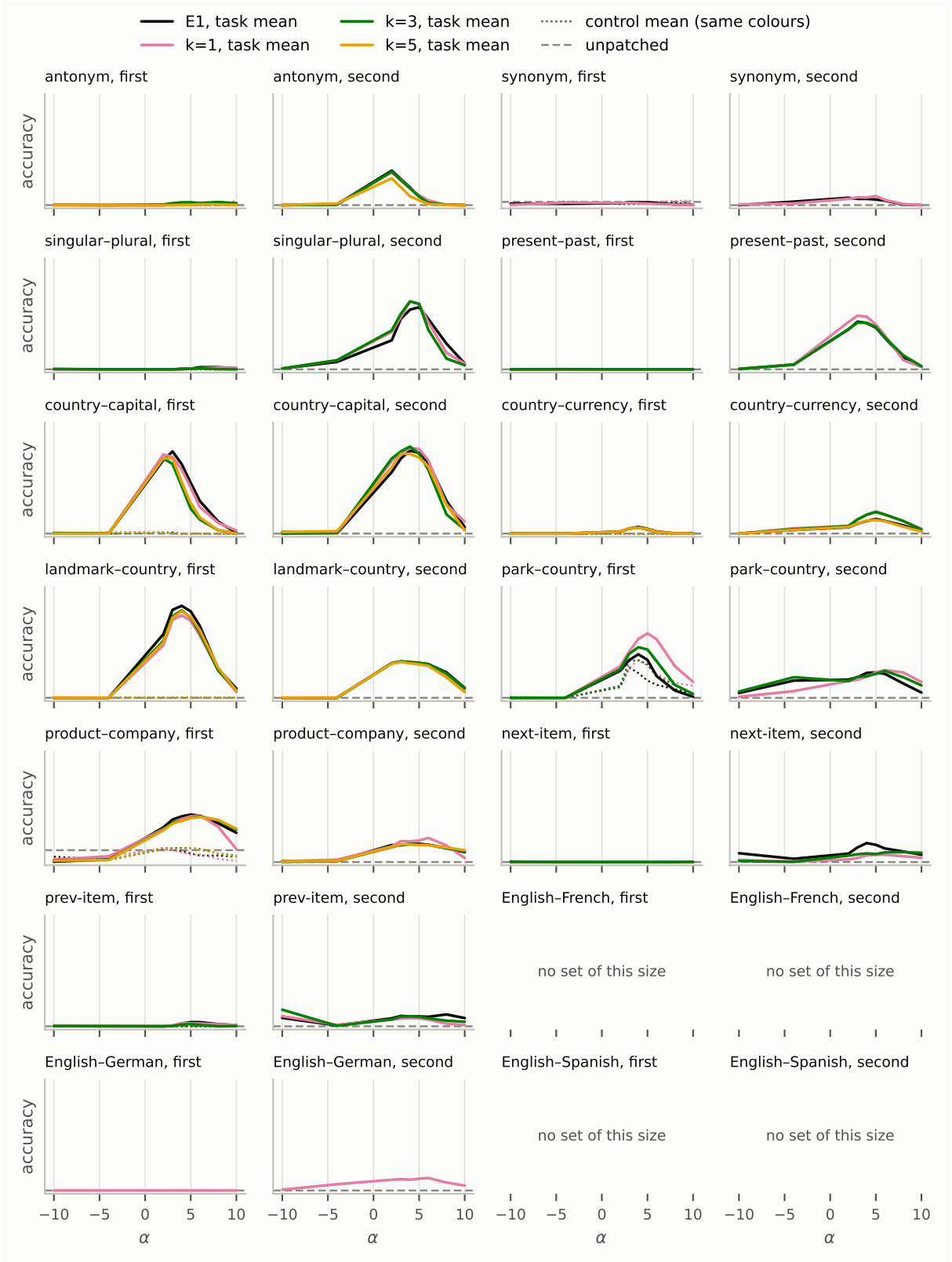


Figure 23: Equal-size sets of 20 Logit-head heads, rule style, E1 and E1b rules, as Figure 21.

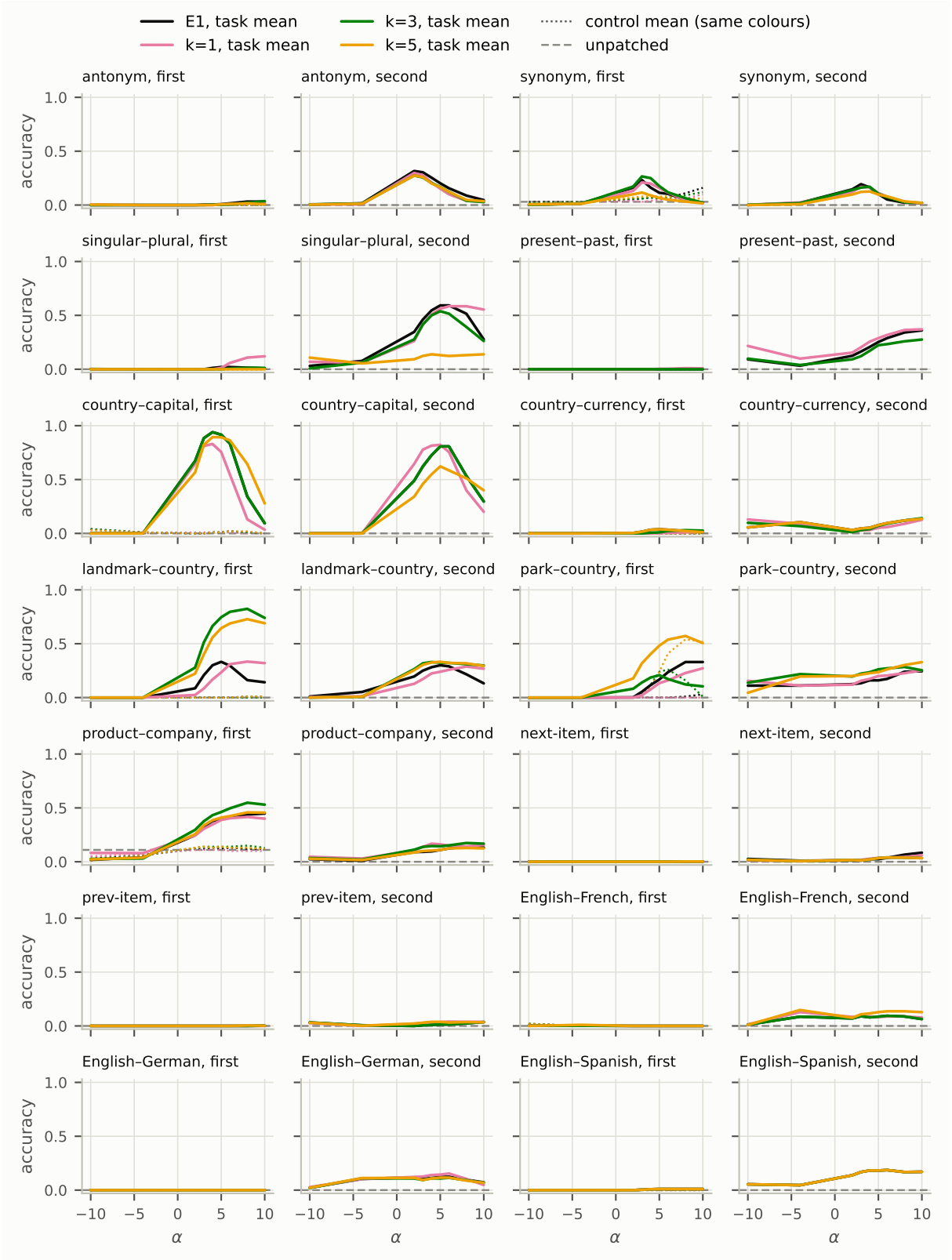


Figure 24: Equal-size sets of 5 J-head heads, rule style, E1 and E1b rules, as Figure 21.

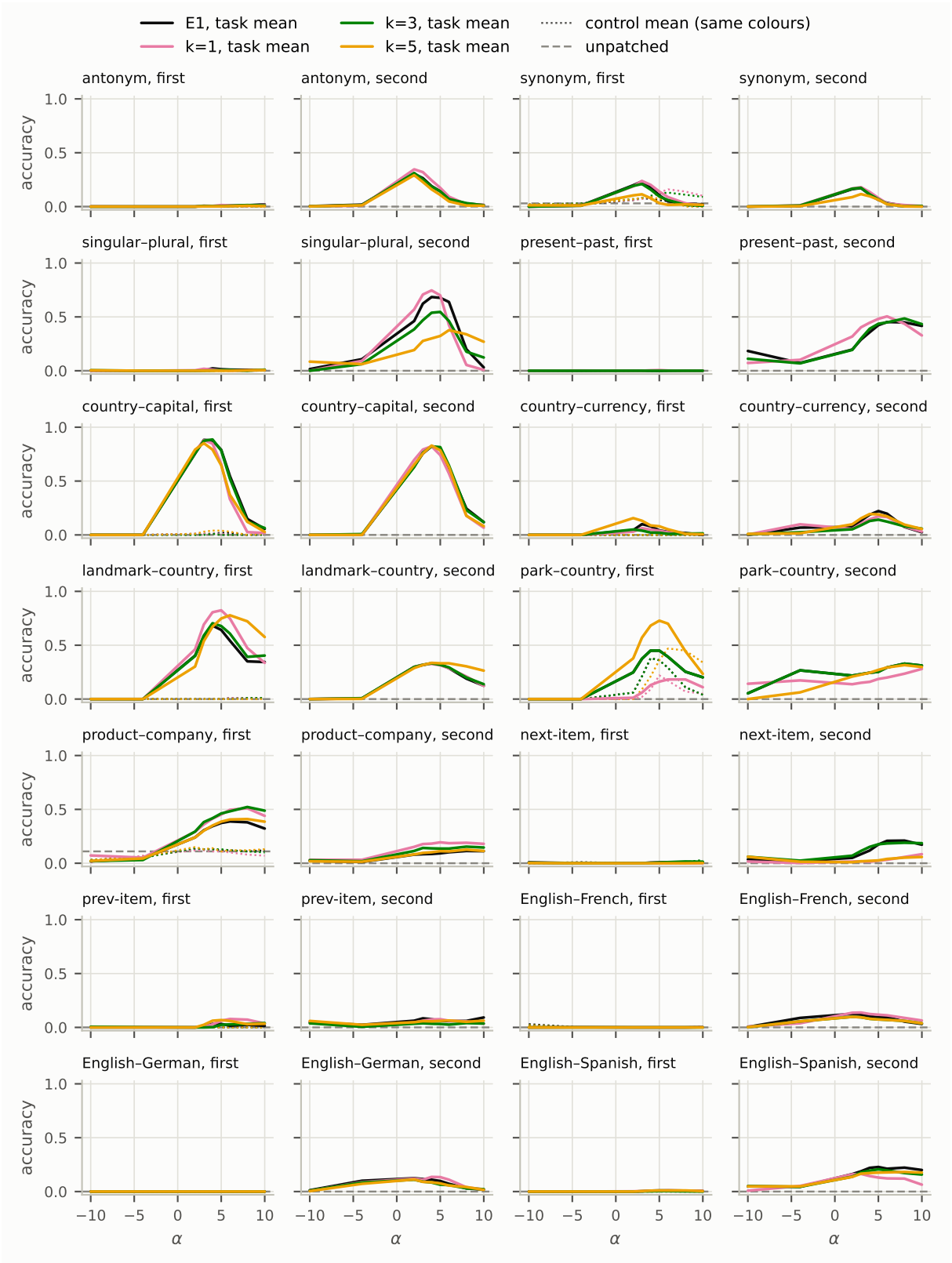


Figure 25: Equal-size sets of 10 J-head heads, rule style, E1 and E1b rules, as Figure 21.

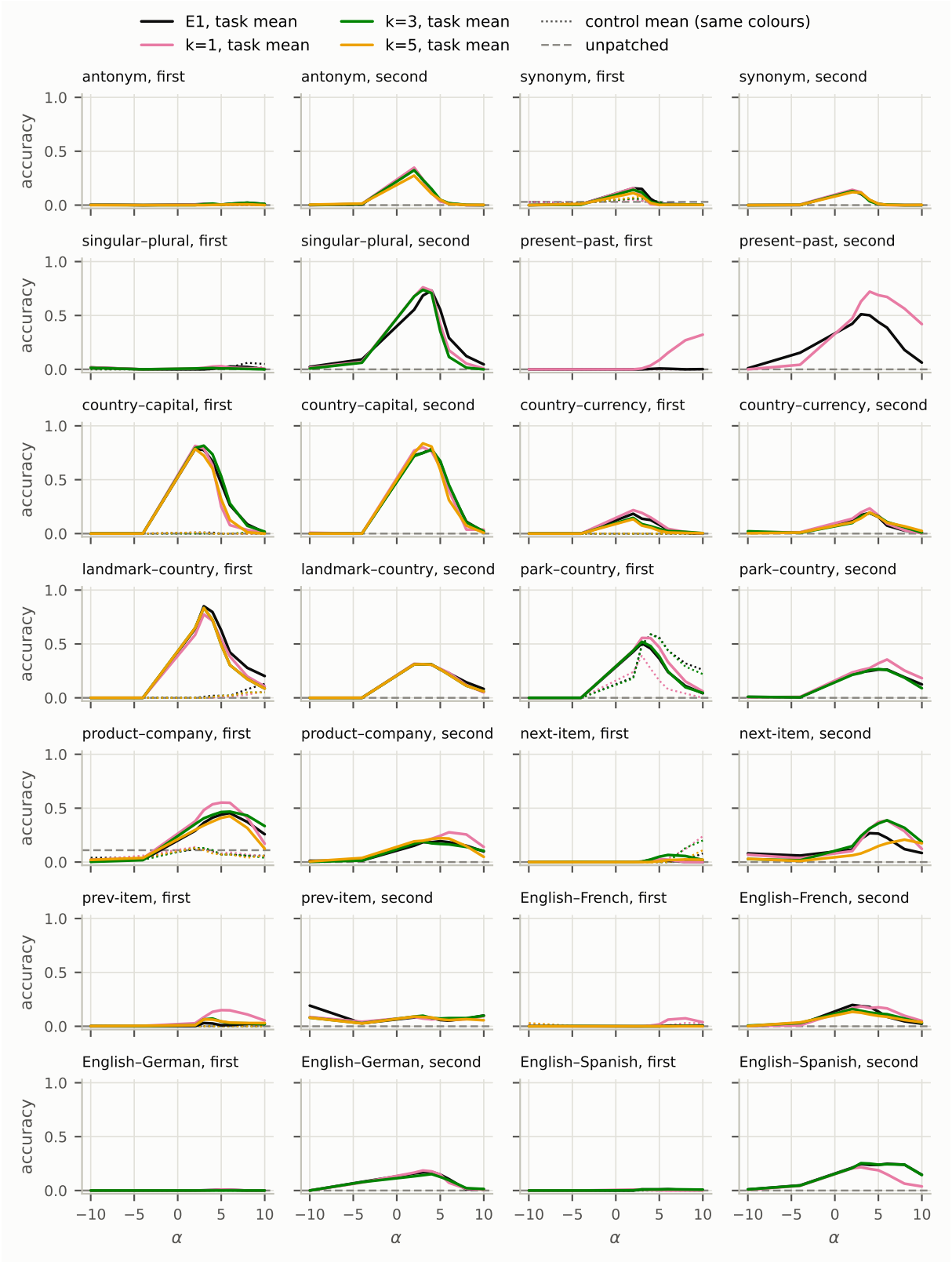


Figure 26: Equal-size sets of 20 J-head heads, rule style, E1 and E1b rules, as Figure 21.

15 Plan

Steps.

1. Run E0 and save every reply. Score the saved outputs with the strict and the flexible score (done) and with the judged score (to run) (Sec. 4.5).
2. Run E1 under the flexible score, with both causal checks.
3. Run E1b on the rule-style heads ($k = 5$, H3), and its exploratory sets for $k = 1$ and 3.
4. Run the follow-ups of the union sets (Secs. 8–13) and write them up.
5. Run the equal-size sets (Sec. 14) and write up their results (done).

Reused code and data: the lexical-task-head code of Yang et al. (2026); the Head-JLens code and matrices of Merlo (2026); the task datasets of Todd et al. (2024).

Limitations. One model. Rule-based prompts contain task words, so a head may decode them by copying from the prompt; the copy share measures this but does not decide it. Under J-head, some sites rank tokens highly on unrelated prompts as well (Merlo, 2026, Sec. 3.6), so a larger J-head set is not by itself evidence of more task representation; the causal check is needed. P^+ and the patched items are chosen by the flexible score, which accepts only the reference answer: a prompt answered with another valid answer is counted as wrong. The judged score depends on one LLM judge, which is not checked against hand labels here; Hua et al. (2025) find that LLM judges agree with human annotators on other tasks.

Deliverables. Head sets, measures and causal-check curves for every task, style and readout.

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A Effects of the equal-size sets

The causal effects of the equal-size sets of Sec. 14, as the highest accuracy over the nine values of α minus the unpatched accuracy.

	Rule style						Example style					
	$N = 5$		$N = 10$		$N = 20$		$N = 5$		$N = 10$		$N = 20$	
Task	L	J	L	J	L	J	L	J	L	J	L	J
antonym	0.03	0.03	0.03	0.02	0.03	0.02	0.05	0.06	0.06	0.05	0.65	0.03
synonym	0.01	0.20	0.00	0.20	0.00	0.13	0.06	0.28	0.07	0.33	0.03	0.13
singular–plural	0.12	0.02	0.00	0.02	0.02	0.03	0.01	0.38	0.04	0.43	0.39	0.90
present–past	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.87	0.02	0.92	–	–
country–capital	0.86	0.94	0.78	0.88	0.76	0.78	0.72	0.70	0.71	0.88	0.75	0.99
country–currency	0.06	0.04	0.06	0.10	0.06	0.18	0.14	0.15	0.08	0.22	0.09	0.21
landmark–country	0.76	0.33	0.91	0.68	0.85	0.85	0.35	0.40	0.67	0.49	0.22	0.36
park–country	0.33	0.33	0.50	0.45	0.40	0.51	0.63	0.45	0.68	0.48	0.48	0.68
product–company	0.34	0.34	0.35	0.28	0.33	0.34	0.27	0.27	0.33	0.14	0.13	0.29
next-item	0.00	0.00	0.00	0.01	0.00	0.02	0.00	0.04	0.32	0.25	0.56	0.58
prev-item	0.00	0.00	0.00	0.03	0.04	0.03	0.00	0.00	0.00	0.00	0.00	0.14
English–French	0.00	0.00	0.00	0.00	–	0.01	0.00	0.00	0.00	0.00	–	0.13
English–German	0.00	0.00	0.00	0.00	–	0.01	0.00	0.00	0.00	0.00	–	0.00
English–Spanish	0.00	0.01	–	0.01	–	0.01	0.00	0.01	–	0.01	–	0.01

Table 14: Equal-size sets, E1 rule, first causal check: causal effect of the N highest-ranked heads with means of their own style, as in Table 5, over the nine values of α. L: Logit-head, J: J-head. –: set not formed (fewer than N heads match on any prompt).

	Rule style						Example style					
	$N = 5$		$N = 10$		$N = 20$		$N = 5$		$N = 10$		$N = 20$	
Task	L	J	L	J	L	J	L	J	L	J	L	J
antonym	0.30	0.32	0.32	0.30	0.32	0.34	0.31	0.31	0.33	0.31	0.42	0.37
synonym	0.05	0.19	0.06	0.17	0.07	0.14	0.05	0.09	0.07	0.13	0.09	0.11
singular–plural	0.58	0.59	0.45	0.68	0.58	0.73	0.69	0.76	0.81	0.86	0.79	0.87
present–past	0.34	0.36	0.43	0.45	0.44	0.51	0.61	0.80	0.56	0.83	–	–
country–capital	0.70	0.81	0.70	0.82	0.77	0.79	0.65	0.53	0.73	0.56	0.70	0.77
country–currency	0.18	0.14	0.20	0.22	0.14	0.19	0.24	0.25	0.29	0.19	0.17	0.21
landmark–country	0.32	0.30	0.33	0.33	0.34	0.31	0.33	0.30	0.32	0.33	0.33	0.32
park–country	0.24	0.24	0.34	0.33	0.24	0.26	0.36	0.36	0.35	0.36	0.45	0.40
product–company	0.14	0.14	0.16	0.11	0.17	0.19	0.12	0.16	0.20	0.23	0.25	0.34
next-item	0.03	0.08	0.07	0.21	0.18	0.27	0.10	0.56	0.45	0.70	0.65	0.81
prev-item	0.04	0.04	0.05	0.09	0.11	0.19	0.12	0.01	0.17	0.19	0.33	0.80
English–French	0.08	0.09	0.10	0.13	–	0.20	0.12	0.13	0.13	0.15	–	0.34
English–German	0.08	0.12	0.10	0.12	–	0.17	0.13	0.09	0.15	0.13	–	0.14
English–Spanish	0.11	0.19	–	0.23	–	0.25	0.06	0.15	–	0.15	–	0.14

Table 15: Equal-size sets, E1 rule, second causal check, as Table 14. Items: the wrongly answered 1-shot prompts patched; the baseline is 0.

	Rule style						Example style					
	$N = 5$		$N = 10$		$N = 20$		$N = 5$		$N = 10$		$N = 20$	
Task	L	J	L	J	L	J	L	J	L	J	L	J
antonym	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.19	0.00
synonym	0.01	0.13	0.01	0.05	0.00	0.03	0.05	0.03	0.05	0.07	0.01	0.05
singular–plural	0.00	0.00	0.01	0.01	0.00	0.06	0.01	0.55	0.01	0.21	0.05	0.04
present–past	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	–	–
country–capital	0.02	0.04	0.01	0.02	0.01	0.01	0.02	0.03	0.02	0.03	0.01	0.02
country–currency	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
landmark–country	0.00	0.01	0.01	0.01	0.00	0.13	0.01	0.00	0.00	0.01	0.00	0.00
park–country	0.03	0.03	0.56	0.38	0.28	0.59	0.57	0.29	0.36	0.19	0.21	0.48
product–company	0.02	0.02	0.03	0.02	0.02	0.01	0.02	0.02	0.01	0.00	0.01	0.03
next-item	0.00	0.00	0.00	0.01	0.00	0.08	0.00	0.01	0.00	0.22	0.08	0.52
prev-item	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
English–French	0.01	0.02	0.01	0.03	–	0.00	0.01	0.01	0.00	0.00	–	0.02
English–German	0.00	0.00	0.00	0.00	–	0.00	0.00	0.00	0.00	0.00	–	0.00
English–Spanish	0.00	0.00	–	0.00	–	0.00	0.00	0.00	–	0.00	–	0.00

Table 16: Equal-size sets, E1 rule, first causal check: causal effect of the control mean at the sets of Table 14.

	$N = 5$				$N = 10$				$N = 20$			
Task	E1	$k = 1$	$k = 3$	$k = 5$	E1	$k = 1$	$k = 3$	$k = 5$	E1	$k = 1$	$k = 3$	$k = 5$
Logit-head												
antonym	0.03	0.01	0.02	0.02	0.03	0.02	0.03	0.02	0.03	0.02	0.03	0.01
synonym	0.01	0.00	0.02	0.01	0.00	0.00	0.00	–	0.00	−0.01	–	–
singular–plural	0.12	0.15	0.17	0.01	0.00	0.02	0.01	0.00	0.02	0.02	0.01	–
present–past	0.00	0.00	0.00	0.00	0.00	0.00	0.00	–	0.00	0.00	0.00	–
country–capital	0.86	0.87	0.84	0.86	0.78	0.81	0.75	0.83	0.76	0.73	0.70	0.70
country–currency	0.06	0.04	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.05
landmark–country	0.76	0.33	0.79	0.72	0.91	0.91	0.91	0.85	0.85	0.76	0.81	0.80
park–country	0.33	0.02	0.57	0.70	0.50	0.68	0.50	0.67	0.40	0.60	0.47	–
product–company	0.34	0.39	0.36	0.23	0.35	0.36	0.28	0.29	0.33	0.32	0.31	0.31
next-item	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	–
prev-item	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.03	0.02	–
English–French	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	–	–	–	–
English–German	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	–	0.00	–	–
English–Spanish	0.00	0.00	0.00	0.00	–	0.00	–	–	–	–	–	–
J-head												
antonym	0.03	0.02	0.04	0.02	0.02	0.02	0.01	0.01	0.02	0.01	0.02	0.01
synonym	0.20	0.18	0.24	0.09	0.20	0.21	0.18	0.08	0.13	0.13	0.11	0.08
singular–plural	0.02	0.12	0.01	0.00	0.02	0.02	0.01	0.01	0.03	0.03	0.01	–
present–past	0.00	0.01	0.00	–	0.00	0.01	0.00	–	0.01	0.32	–	–
country–capital	0.94	0.83	0.94	0.89	0.88	0.88	0.89	0.85	0.78	0.81	0.82	0.79
country–currency	0.04	0.02	0.03	0.04	0.10	0.06	0.05	0.16	0.18	0.22	0.15	0.13
landmark–country	0.33	0.33	0.82	0.73	0.68	0.82	0.70	0.78	0.85	0.77	0.83	0.83
park–country	0.33	0.27	0.21	0.57	0.45	0.18	0.45	0.73	0.51	0.56	0.52	–
product–company	0.34	0.31	0.44	0.35	0.28	0.40	0.41	0.30	0.34	0.44	0.36	0.32
next-item	0.00	0.00	0.00	0.00	0.01	0.00	0.02	0.00	0.02	0.03	0.07	0.02
prev-item	0.00	0.00	0.00	0.00	0.03	0.08	0.04	0.07	0.03	0.15	0.07	0.06
English–French	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.00	0.01	0.07	0.00	0.00
English–German	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.00	–
English–Spanish	0.01	0.01	0.01	0.01	0.01	0.01	0.00	0.01	0.01	0.01	0.01	–

Table 17: Equal-size sets, rule style, first causal check: causal effect of the N highest-ranked heads under the rule of E1 and the rules of E1b (at least k of the five templates), as Table 14.

	$N = 5$				$N = 10$				$N = 20$			
Task	E1	$k = 1$	$k = 3$	$k = 5$	E1	$k = 1$	$k = 3$	$k = 5$	E1	$k = 1$	$k = 3$	$k = 5$
Logit-head												
antonym	0.30	0.27	0.26	0.28	0.32	0.28	0.29	0.29	0.32	0.30	0.31	0.25
synonym	0.05	0.04	0.04	0.06	0.06	0.04	0.06	–	0.07	0.08	–	–
singular–plural	0.58	0.60	0.62	0.32	0.45	0.49	0.52	0.35	0.58	0.63	0.63	–
present–past	0.34	0.42	0.38	0.39	0.43	0.40	0.40	–	0.44	0.50	0.43	–
country–capital	0.70	0.81	0.73	0.72	0.70	0.81	0.75	0.70	0.77	0.79	0.81	0.74
country–currency	0.18	0.14	0.18	0.18	0.20	0.19	0.18	0.15	0.14	0.21	0.20	0.13
landmark–country	0.32	0.29	0.32	0.33	0.33	0.33	0.33	0.34	0.34	0.34	0.34	0.33
park–country	0.24	0.19	0.33	0.38	0.34	0.37	0.34	0.32	0.24	0.26	0.25	–
product–company	0.14	0.12	0.15	0.12	0.16	0.15	0.11	0.17	0.17	0.22	0.17	0.16
next-item	0.03	0.04	0.04	0.04	0.07	0.08	0.06	0.07	0.18	0.07	0.09	–
prev-item	0.04	0.02	0.03	0.04	0.05	0.03	0.04	0.05	0.11	0.09	0.15	–
English–French	0.08	0.09	0.09	0.08	0.10	0.10	0.10	0.10	–	–	–	–
English–German	0.08	0.08	0.08	0.08	0.10	0.09	0.09	0.10	–	0.12	–	–
English–Spanish	0.11	0.14	0.11	0.11	–	0.15	–	–	–	–	–	–
J-head												
antonym	0.32	0.30	0.27	0.27	0.30	0.35	0.31	0.29	0.34	0.35	0.32	0.27
synonym	0.19	0.17	0.17	0.13	0.17	0.18	0.17	0.12	0.14	0.14	0.13	0.12
singular–plural	0.59	0.58	0.54	0.14	0.68	0.75	0.55	0.38	0.73	0.76	0.74	–
present–past	0.36	0.37	0.28	–	0.45	0.50	0.49	–	0.51	0.72	–	–
country–capital	0.81	0.82	0.81	0.62	0.82	0.82	0.82	0.83	0.79	0.80	0.78	0.84
country–currency	0.14	0.13	0.14	0.14	0.22	0.17	0.14	0.19	0.19	0.23	0.19	0.20
landmark–country	0.30	0.29	0.33	0.33	0.33	0.33	0.33	0.34	0.31	0.32	0.31	0.31
park–country	0.24	0.25	0.28	0.33	0.33	0.28	0.33	0.32	0.26	0.36	0.27	–
product–company	0.14	0.17	0.17	0.13	0.11	0.19	0.15	0.13	0.19	0.28	0.19	0.22
next-item	0.08	0.06	0.04	0.04	0.21	0.08	0.19	0.06	0.27	0.39	0.39	0.21
prev-item	0.04	0.04	0.04	0.04	0.09	0.08	0.04	0.06	0.19	0.09	0.10	0.09
English–French	0.09	0.13	0.09	0.15	0.13	0.14	0.10	0.10	0.20	0.19	0.16	0.13
English–German	0.12	0.15	0.11	0.12	0.12	0.14	0.11	0.11	0.17	0.19	0.15	–
English–Spanish	0.19	0.19	0.19	0.19	0.23	0.16	0.21	0.18	0.25	0.22	0.25	–

Table 18: Equal-size sets, rule style, second causal check, as Table 17. The baseline is 0.

Decision log

2026-10-27 Scope: test the lexical-task-head method of Yang et al. (2026) on complex tasks and compare the per-head logit lens with Head-JLens. New title.

2026-10-27 Tasks: the 14 simple tasks shared by Yang et al. (2026) and Davidson et al. (2025); the five tasks of Fu et al. (2026).

2026-10-27 No main condition: rule-based and example-based prompts are treated alike (replacing the earlier choice of rules as main and examples as secondary).

2026-10-27 Method: lexical task heads, with Logit-head and J-head readouts. Function vectors, function-specific heads and the hidden-state difference between prompts are not used for now.

2026-10-27 Answer format as in the source studies.

2026-10-27 Model: Llama-3.1-8B-Instruct, the main model of Yang et al. (2026) and the only one with released Head-JLens matrices.

2026-10-27 Accuracy is computed with each source paper's evaluation.

2026-10-27 Prompts on which the model does not answer straight away are left out; other ways of handling them are considered after E0.

2026-10-28 The simple tasks follow the code of Yang et al. (2026) exactly, except for the J-head readout and the bfloat16 precision of the Head-JLens matrices: five instruction templates (their code and paper do not use the sixth template of their instruction file); heads from one wording per style (5-shot, and the most accurate template); one mean and one patch per source wording; their control mean; their batching. This replaces, for the simple tasks, the pooling of prompts and means over wordings and the use of six templates. E1 was rerun.

2026-10-28 Simple tasks: the padding of each batch is masked. The code of Yang et al. (2026) passes no attention mask, so the model attends to the padding, which their paper does not describe. E1 was rerun.

2026-10-28 Simple tasks: a second causal check with the setting of the one causal script of Yang et al. (2026) (wrongly answered 1-shot prompts patched, no control). It reproduces their Figs. 57 and 62, which the zero-shot procedure of their App. G.1 does not; both checks are reported.

2026-10-29 Answer evaluation (Sec. 4.5), after Hua et al. (2025): every E0 reply is generated once with longer limits and saved; each is scored with the source evaluation and with a GPT-5.5 judge, and every disagreement is given a reason. The judge sees every reply, not only those the strict score rejects. Head identification and the causal checks keep the strict score until this evaluation is done.

2026-10-29 One generation scheme per group of tasks (Sec. 4.5). Simple tasks: a reply ends at `;`, at a newline once the reply has begun (a leading newline no longer ends it), at the end of sequence, or at 32 new tokens. Complex tasks: a reply ends at the end of sequence or at 1,024 new tokens, for all five tasks; this replaces the limit of 6,000 in reasoning mode and the stop once the scored answer is complete. The saved outputs are generated again under this scheme. Head identification and the causal checks in reasoning mode were run with the earlier limit of 6,000 tokens and the stop at the answer.

2026-10-29 Heads are read at the last prompt token only; the pre-answer position is dropped, with its results. Which prompts form P^+ for the complex tasks is decided later. Causal checks: the patch is applied at the last prompt position only, for every task, and every task uses the same 15 values of α.

2026-10-30 Simple tasks: a second, flexible score next to the first-token score, which is unchanged: the whole reference at the start of the reply, ignoring case, accents, punctuation and a leading article (Sec. 4.5). It is reported for E0; head identification and the causal checks keep the first-token score.

2026-10-30 No hand-labelled validation of the judge; its use rests on the human check of Hua et al. (2025).

2026-10-30 Head identification and the causal checks use the flexible score, replacing the first-token score (simple tasks) and the strict score (complex tasks): P^+ is the prompts answered correctly under it, with no condition that the answer comes first, for both families; the identification wordings, the head sets, the means, the patched items and the controls are chosen by it. Readings and means are taken from the prompts of the saved outputs. In the causal checks the reply is generated as the saved outputs and scored by the flexible score; for the simple tasks this replaces the first token, which is kept as a second score. Both complex-task checks patch 100 items on every task, replacing 10 on Set Game, Operator Function and Tapatan; Operator Function has no second check, as no item is answered wrongly under its original rules prompt. E1 and E2 are rerun.

2026-11-02 E1b: rule-style heads of the simple tasks that meet Definition 1 under all five instruction templates (H3), patched with both causal checks of E1. Only the rule style and only the simple tasks.

2026-11-02 E1b extended after the result of H3, as an exploratory question without a hypothesis: the heads that meet Definition 1 under at least one and at least three of the five templates, patched with both checks. E1b now reports the Logit-head and J-head sets only, not the heads found only by J-head, and no copy share.

2026-11-03 The project is split into two documents: `proposal-simple`, the identification method on the simple tasks (E0 for the simple tasks, E1, E1b), and `proposal-complex`, the complex tasks (E0 for the complex tasks, E2). Work continues on the simple tasks for now. The readout hypothesis is stated once per document, with its criteria unchanged.

2026-11-03 `proposal-simple` is reorganised: the replication of Yang et al. (2026) under both readouts first, then one section per change tried, each with its design followed by its results.

2026-11-03 Head sets of equal size (Sec. 14): the $N \in \{5, 10, 20\}$ highest-ranked heads under each readout and rule, only among heads that match on at least one prompt; patched with own-style means at nine values of α (-10, -4, 2, 3, 4, 5, 6, 8, 10) to shorten the run, with the threshold sets compared over the same nine; identical sets are patched once.